

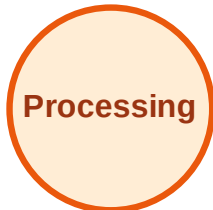
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



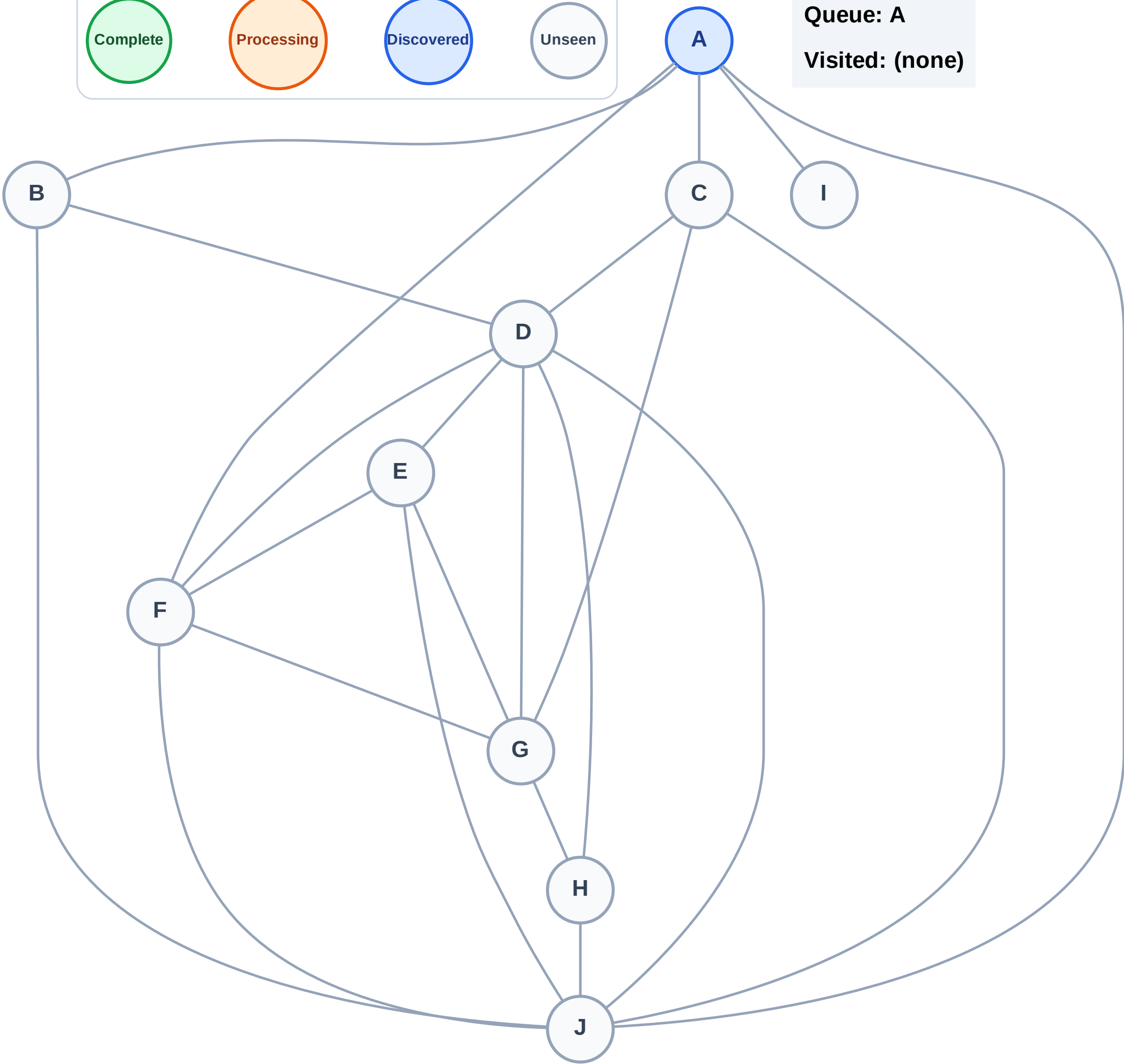
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

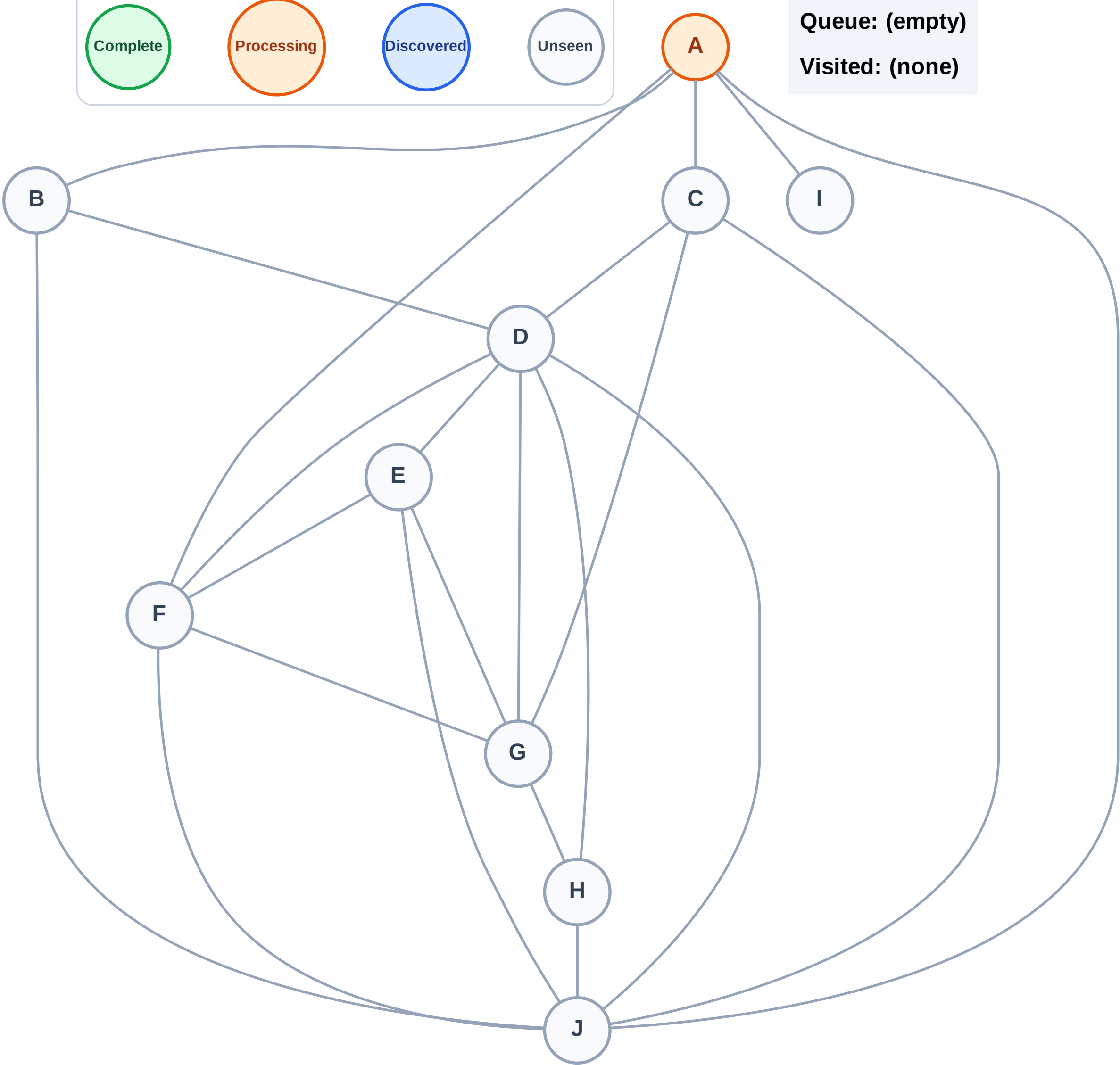
Complete

Processing

Discovered

Unseen

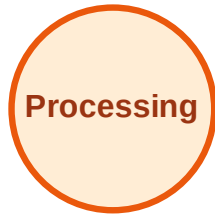
Queue: (empty)
Visited: (none)



BFS Traversal

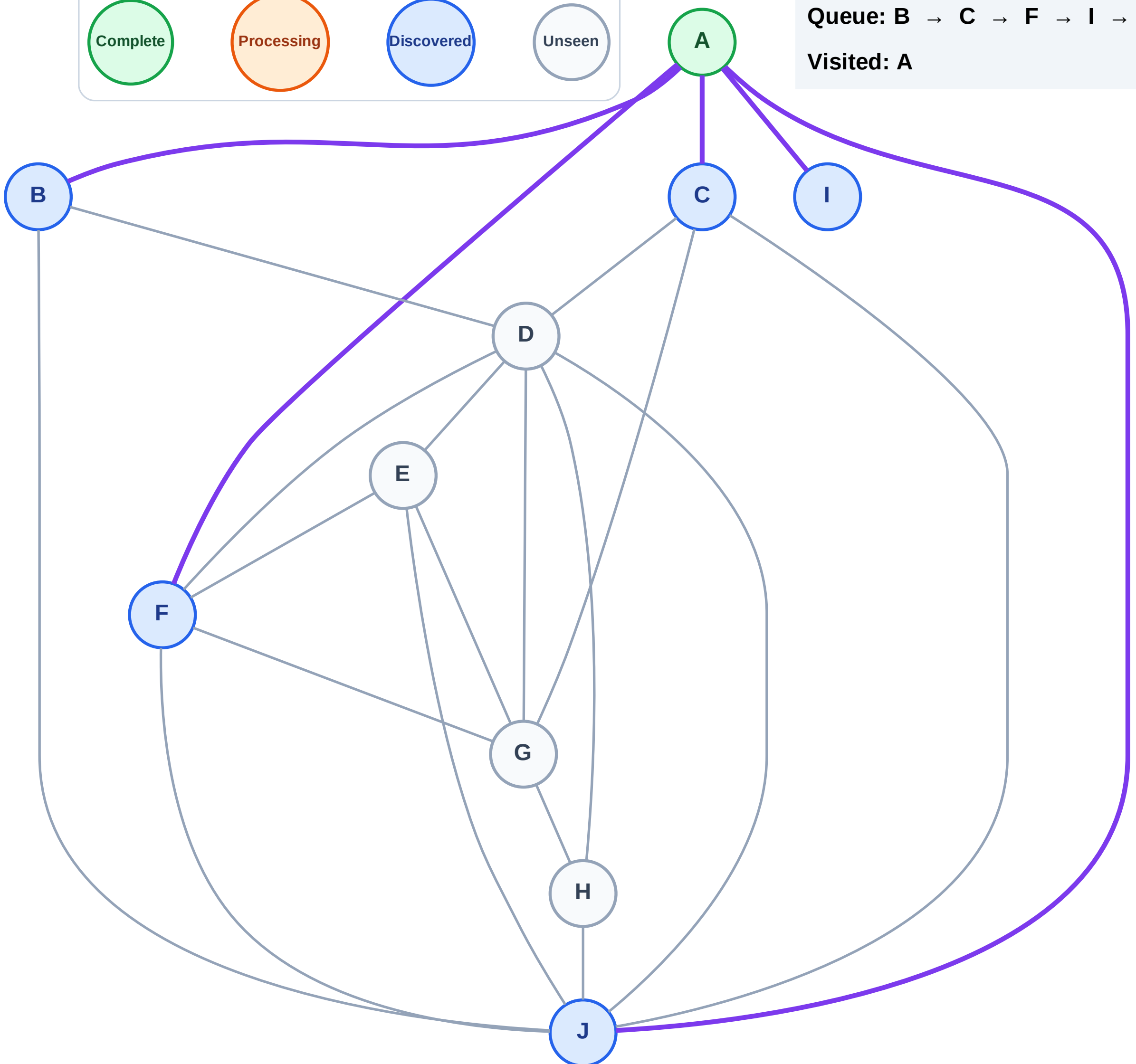
Step 3: Discover B, C, F, I, J from A

Legend



Queue: B → C → F → I → J

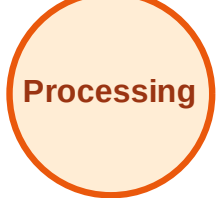
Visited: A



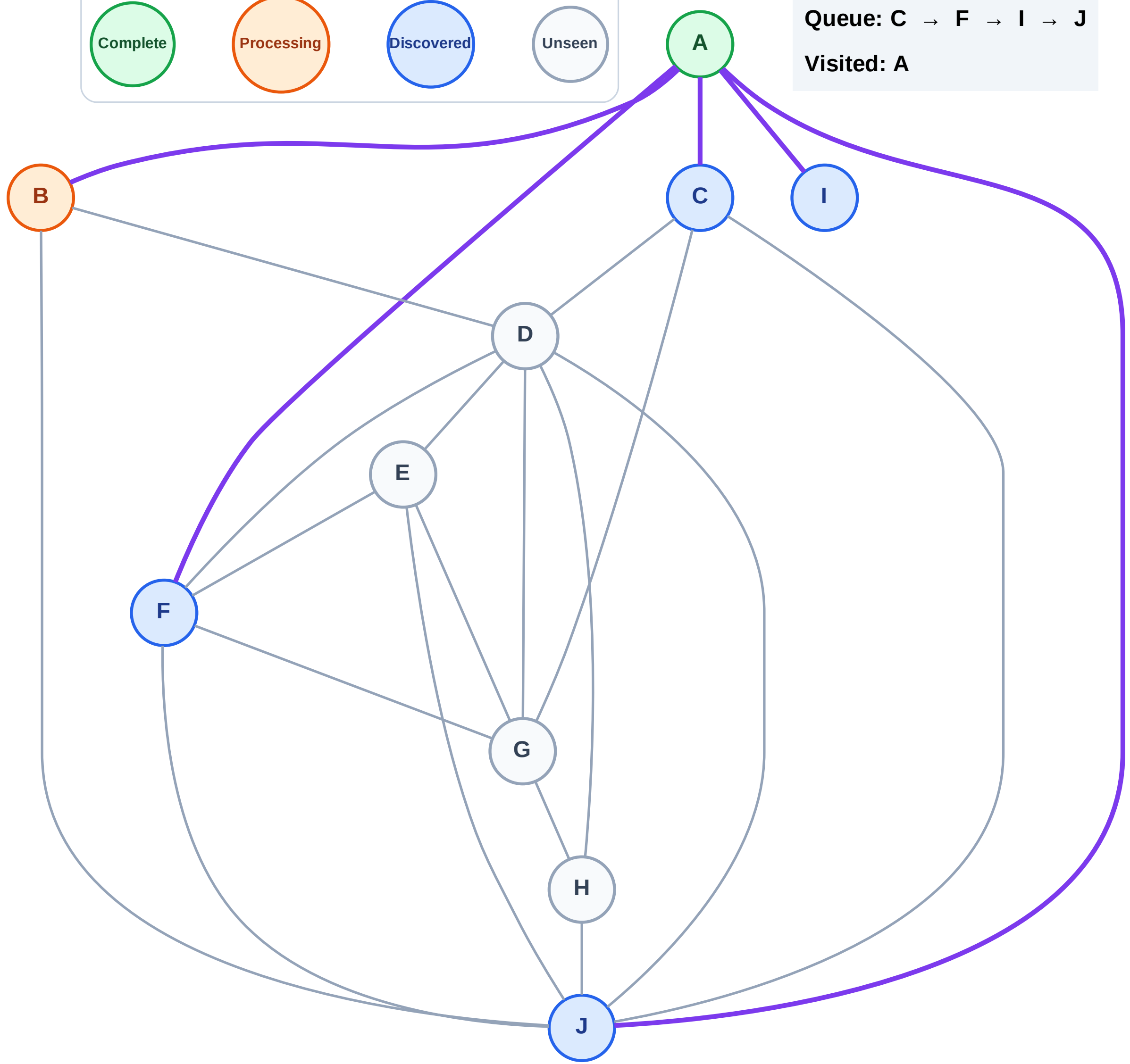
BFS Traversal

Step 4: Process node B

Legend



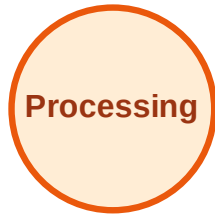
Queue: C → F → I → J
Visited: A



BFS Traversal

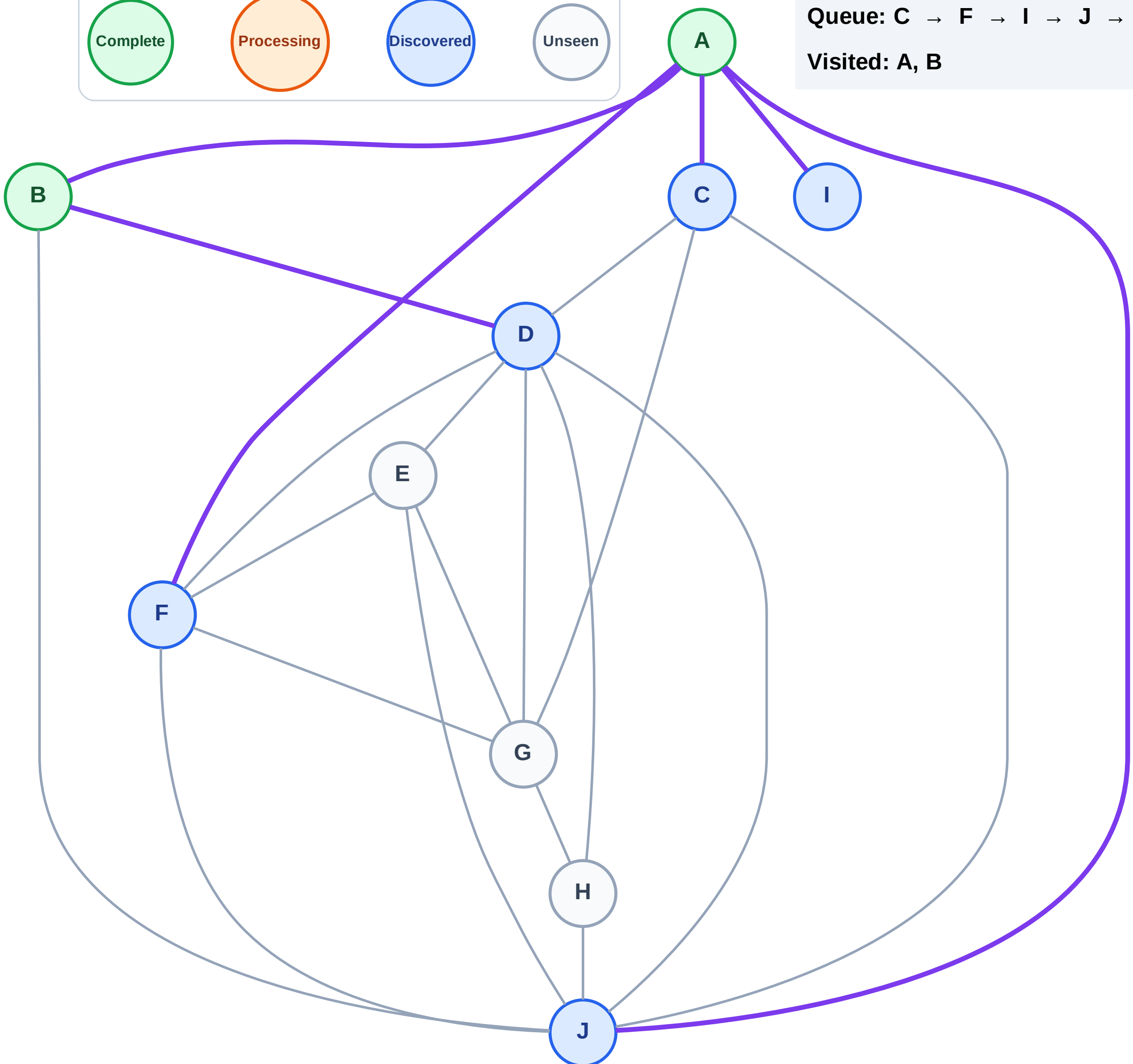
Step 5: Discover D from B

Legend



Queue: C → F → I → J → D

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

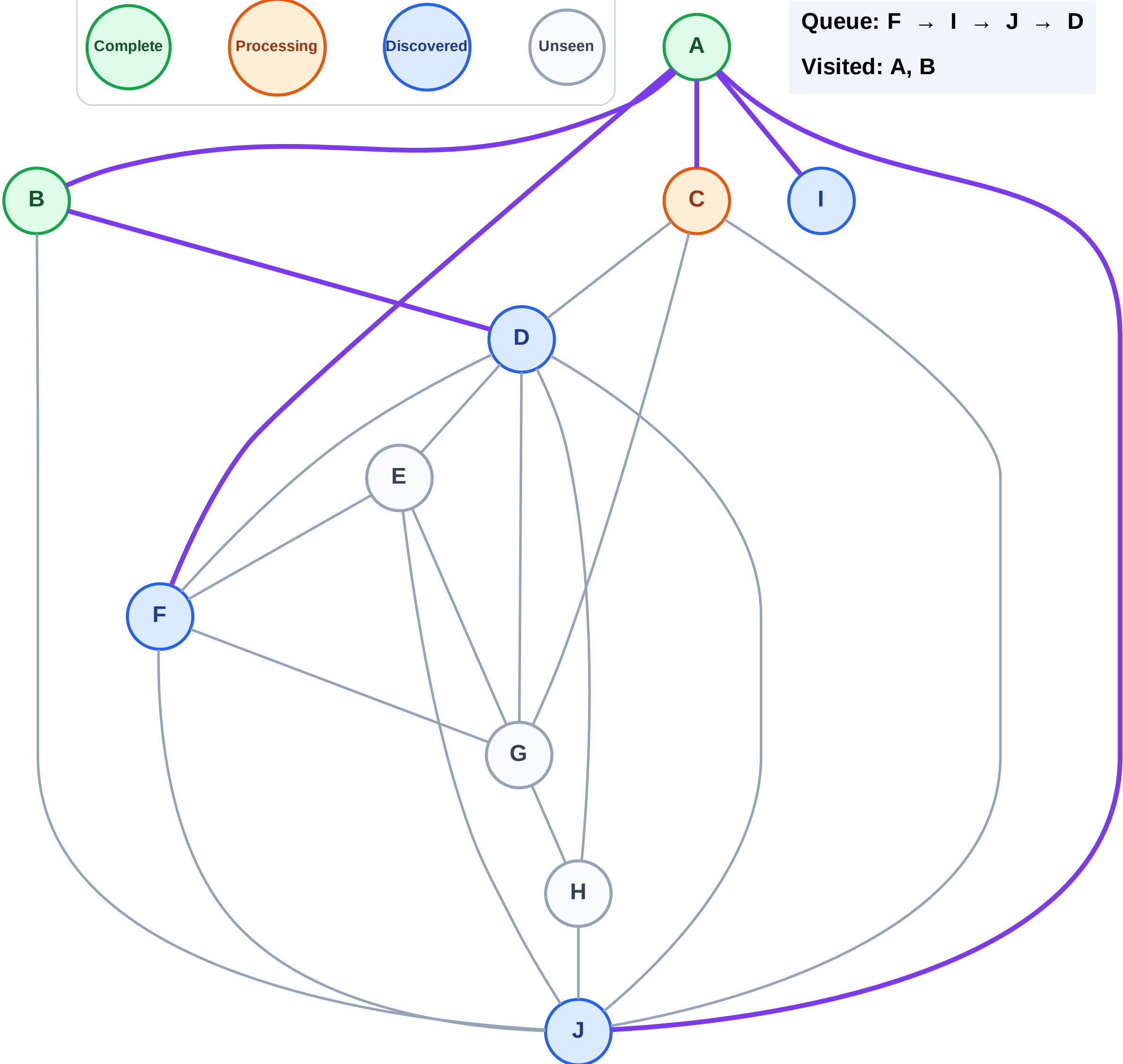
Processing

Discovered

Unseen

Queue: F → I → J → D

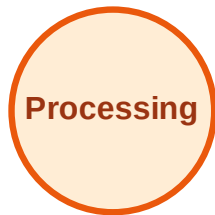
Visited: A, B



BFS Traversal

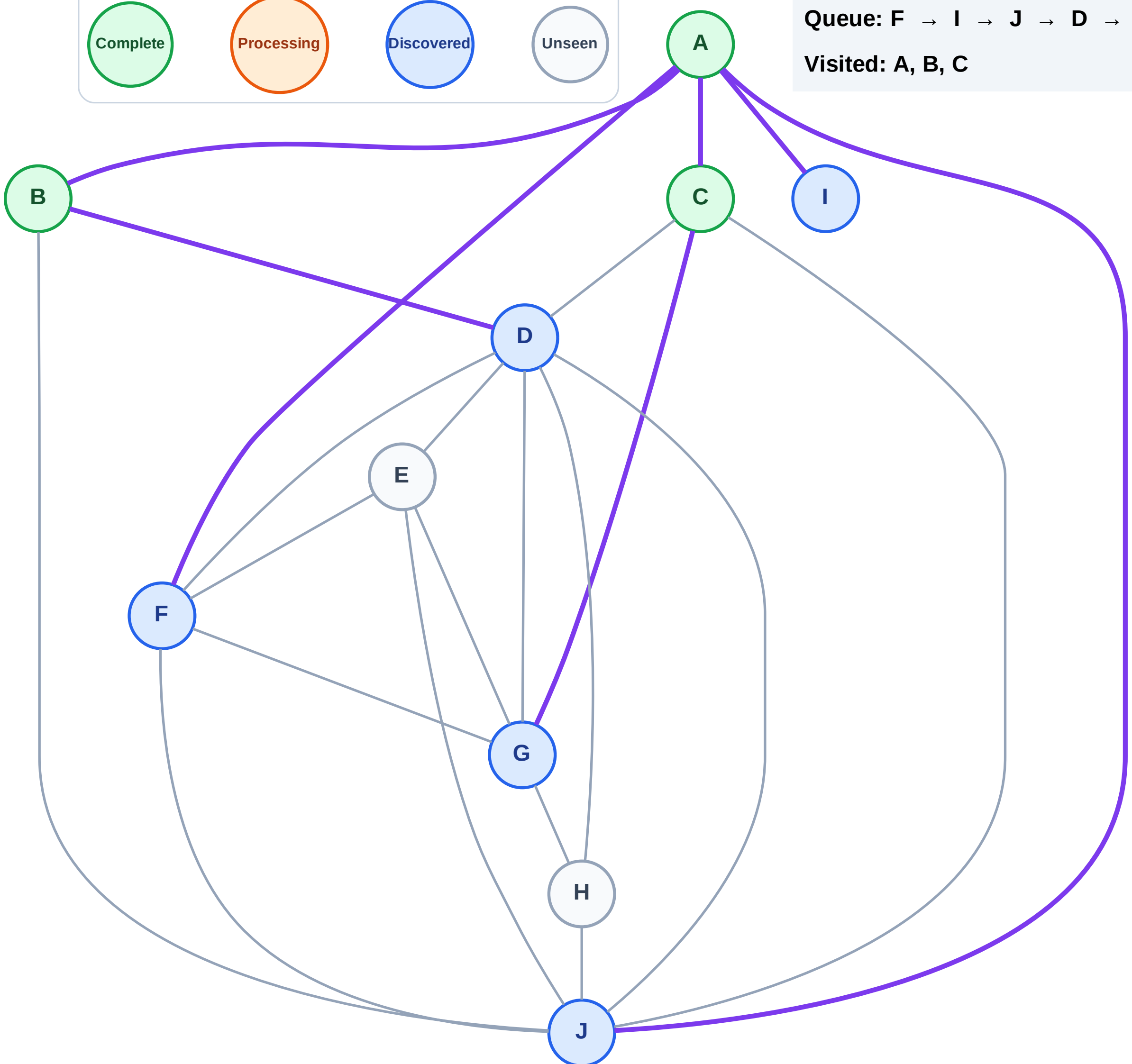
Step 7: Discover G from C

Legend



Queue: F → I → J → D → G

Visited: A, B, C



BFS Traversal

Step 8: Process node F

Legend

Complete

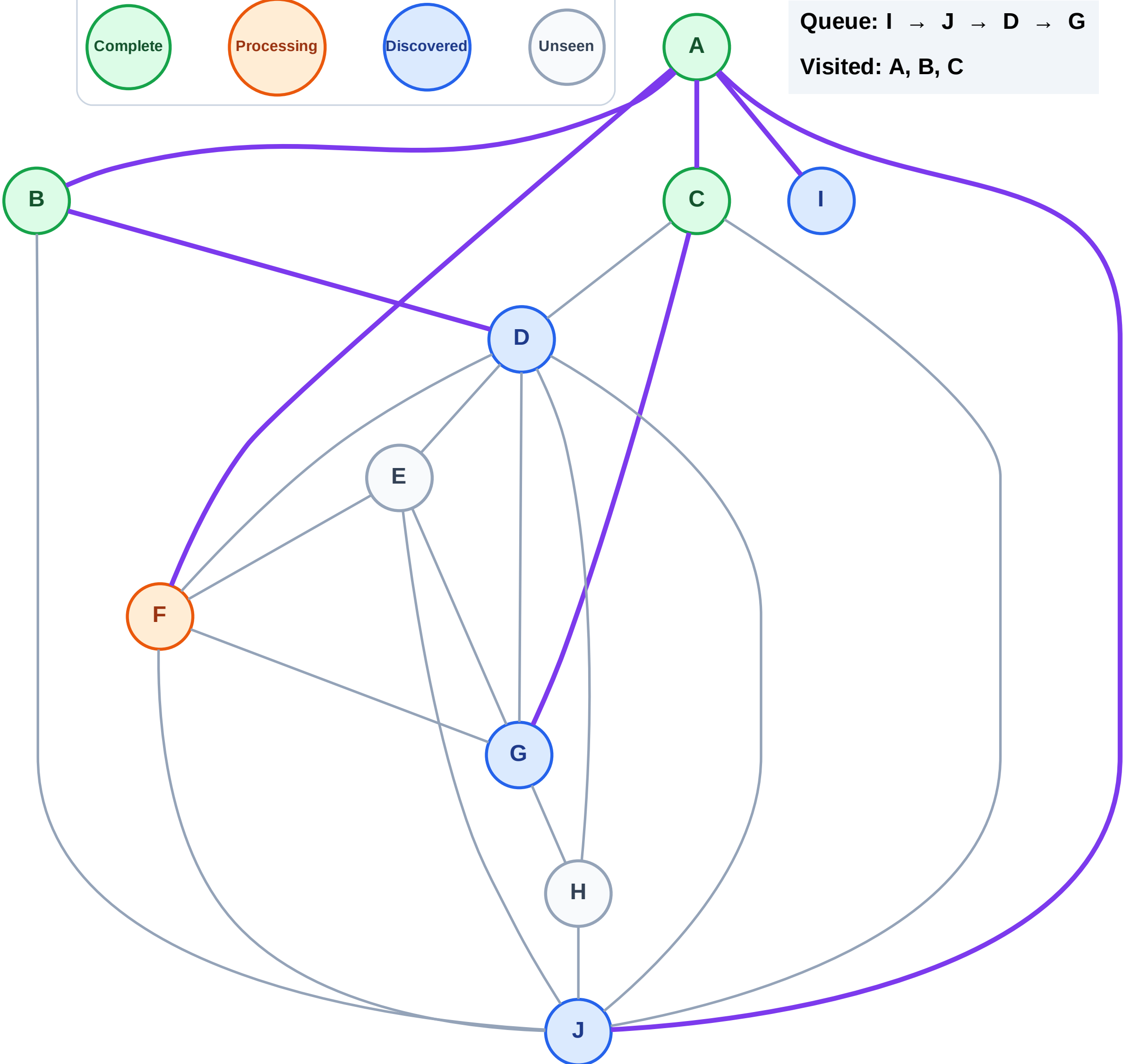
Processing

Discovered

Unseen

Queue: I → J → D → G

Visited: A, B, C



BFS Traversal

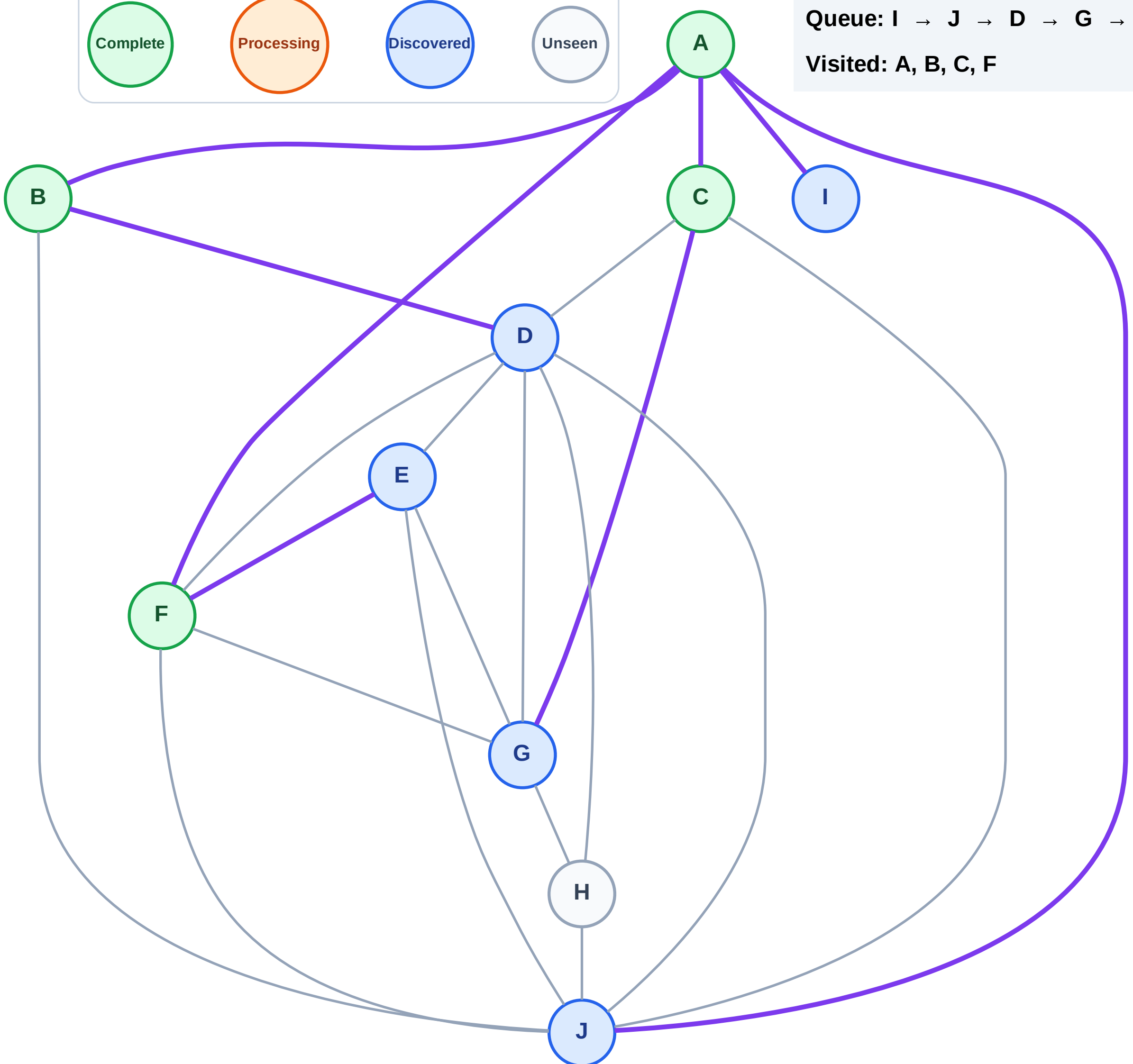
Step 9: Discover E from F

Legend



Queue: I → J → D → G → E

Visited: A, B, C, F



BFS Traversal

Step 10: Process node I

Legend

Complete

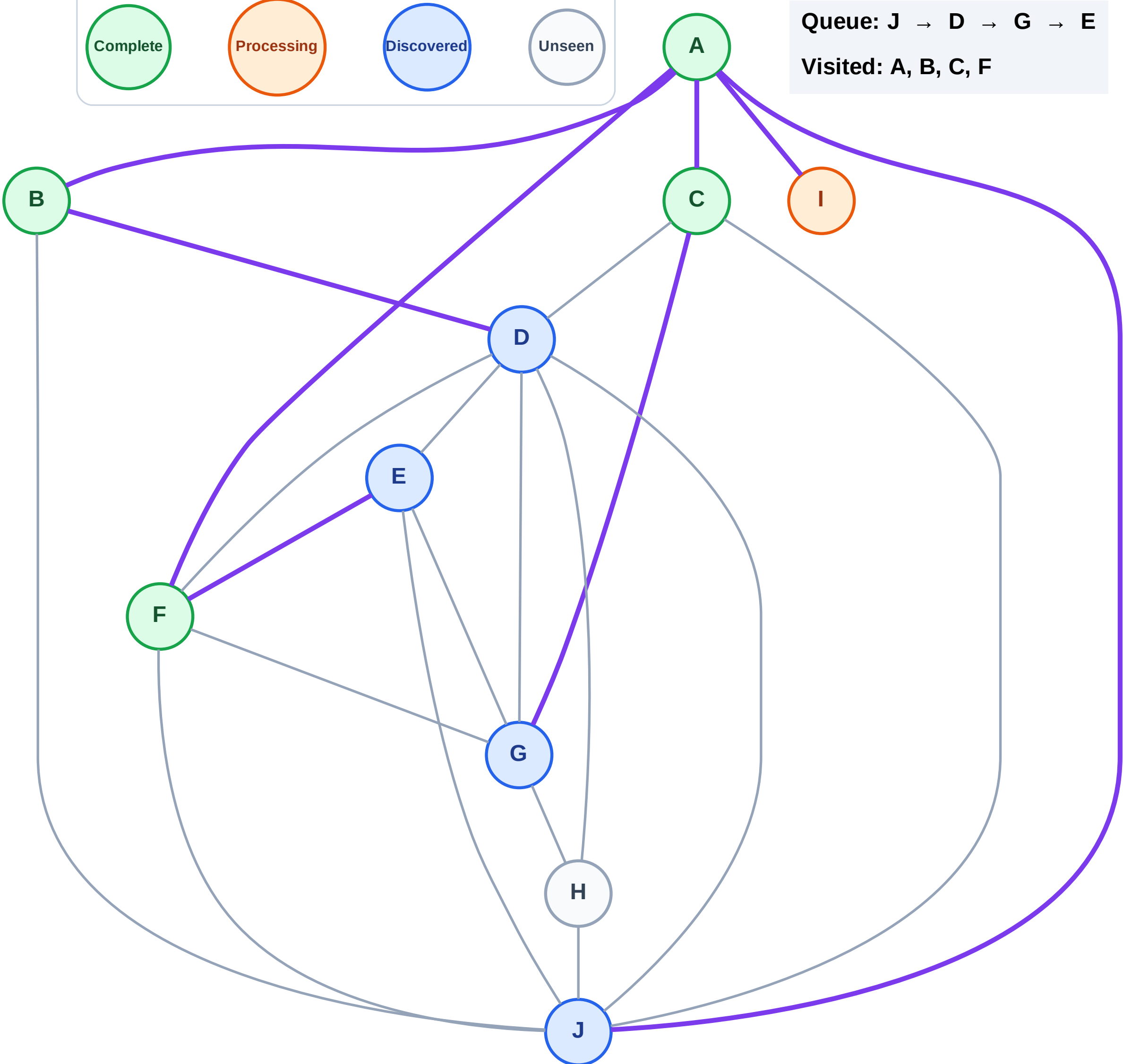
Processing

Discovered

Unseen

Queue: J → D → G → E

Visited: A, B, C, F



BFS Traversal

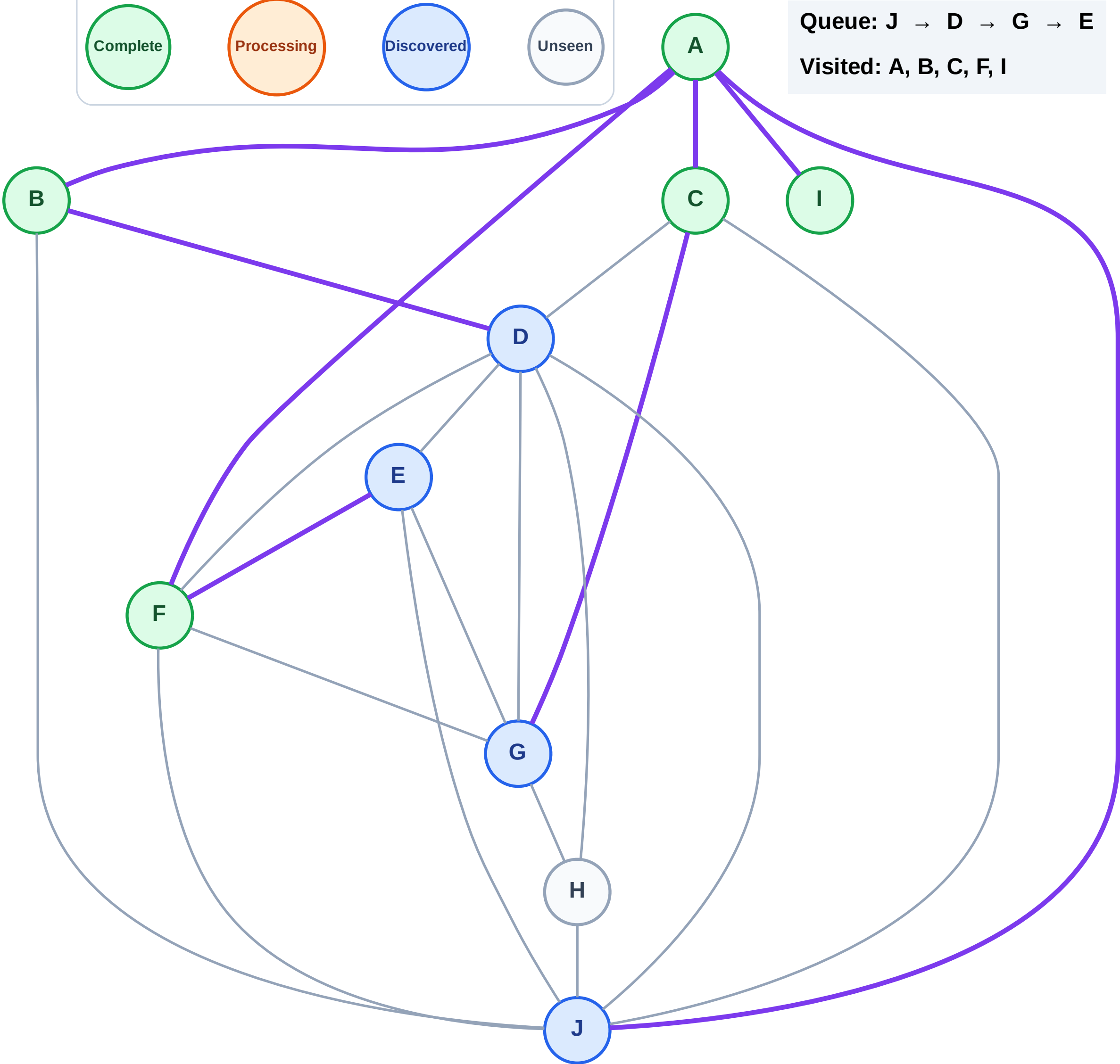
Step 11: No new neighbors from I

Legend



Queue: J → D → G → E

Visited: A, B, C, F, I



BFS Traversal

Step 12: Process node J

Legend

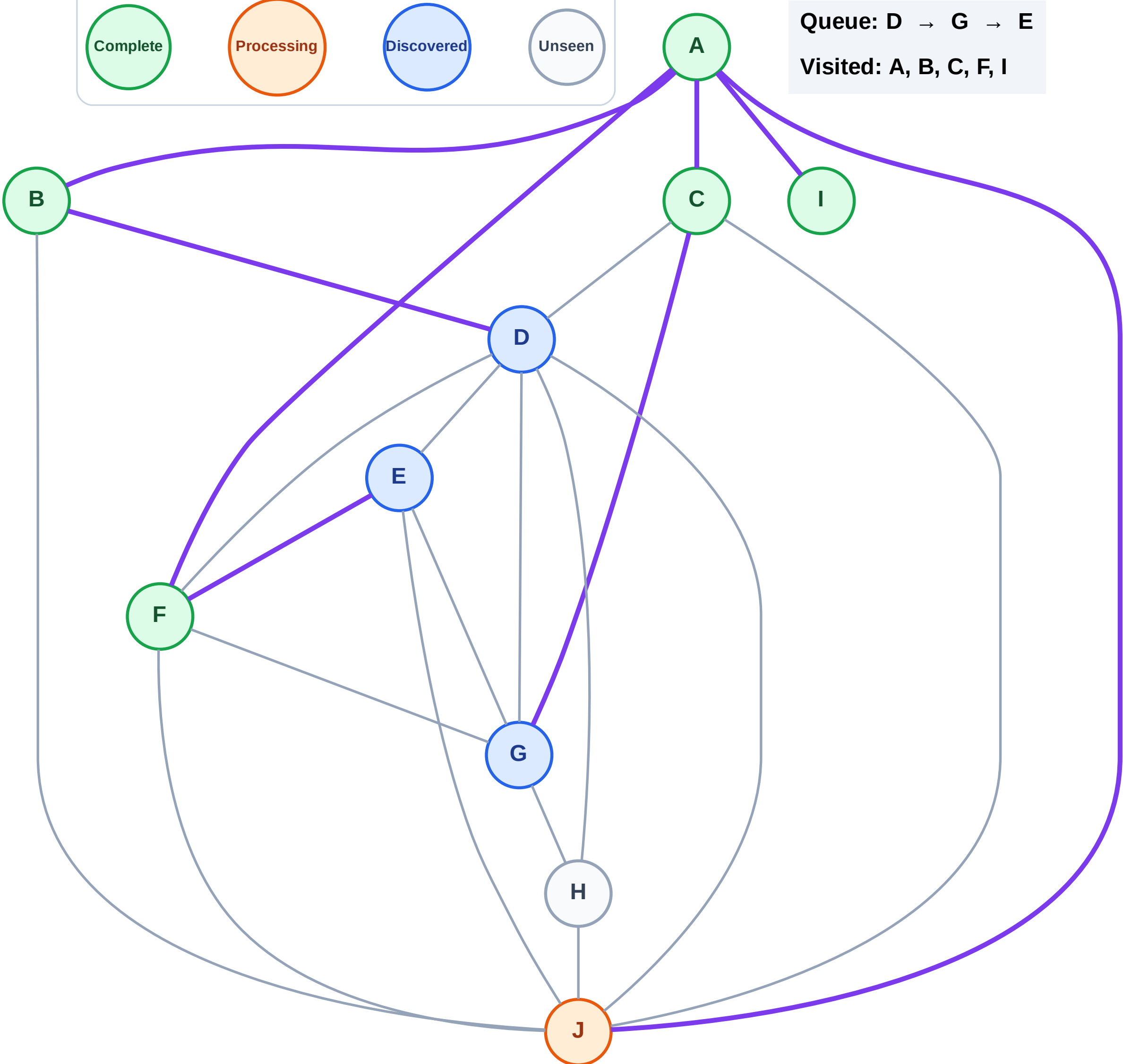
Complete

Processing

Discovered

Unseen

Queue: D → G → E
Visited: A, B, C, F, I



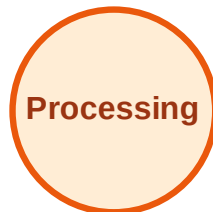
BFS Traversal

Step 13: Discover H from J

Legend



Complete



Processing



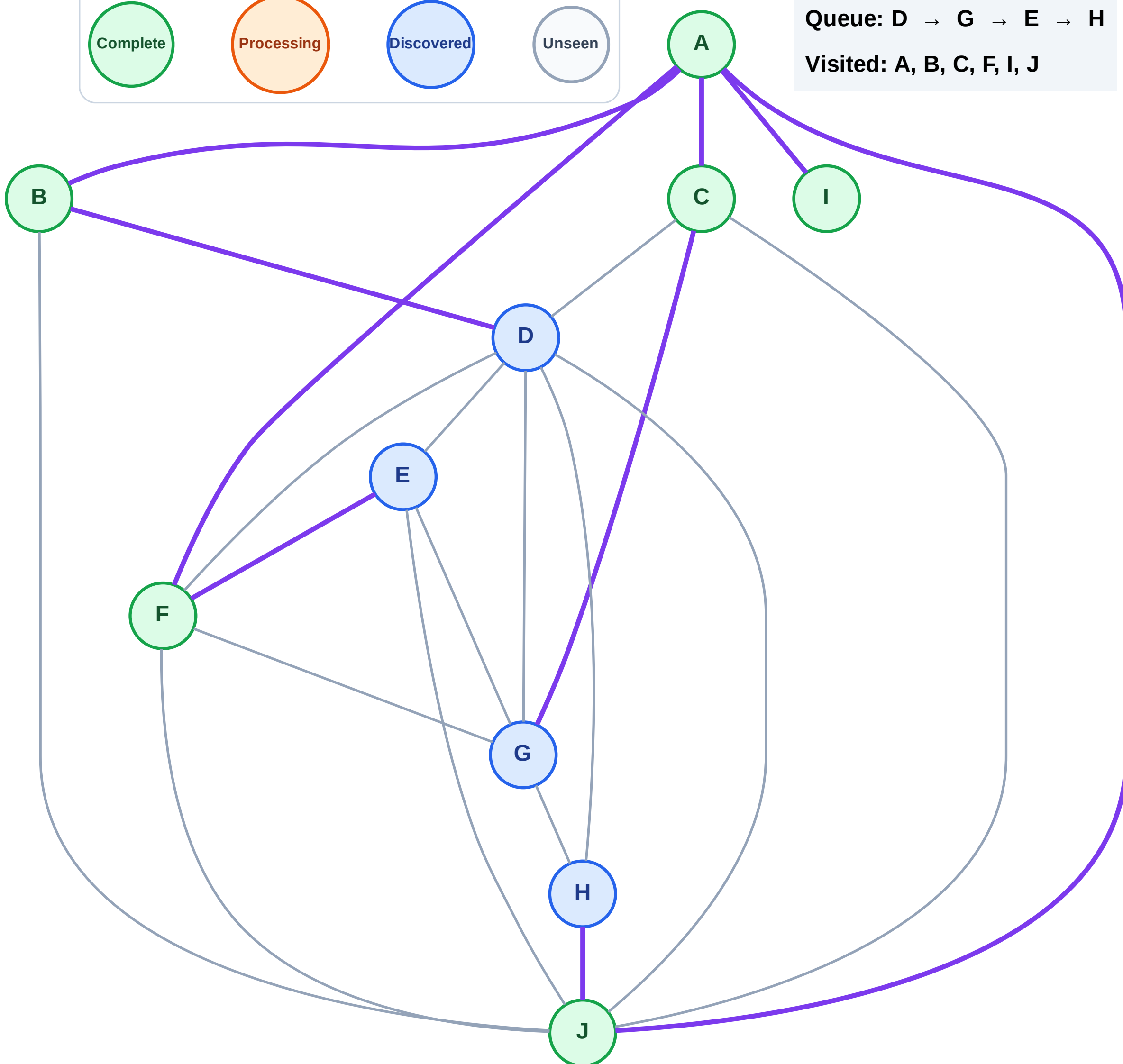
Discovered



Unseen

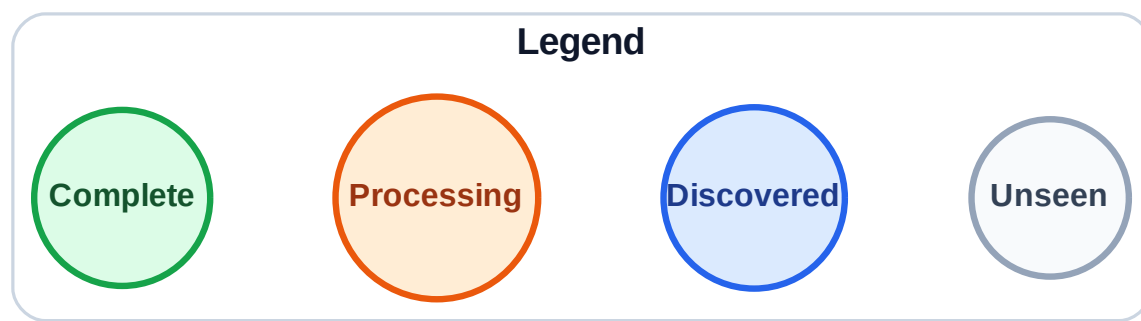
Queue: D → G → E → H

Visited: A, B, C, F, I, J

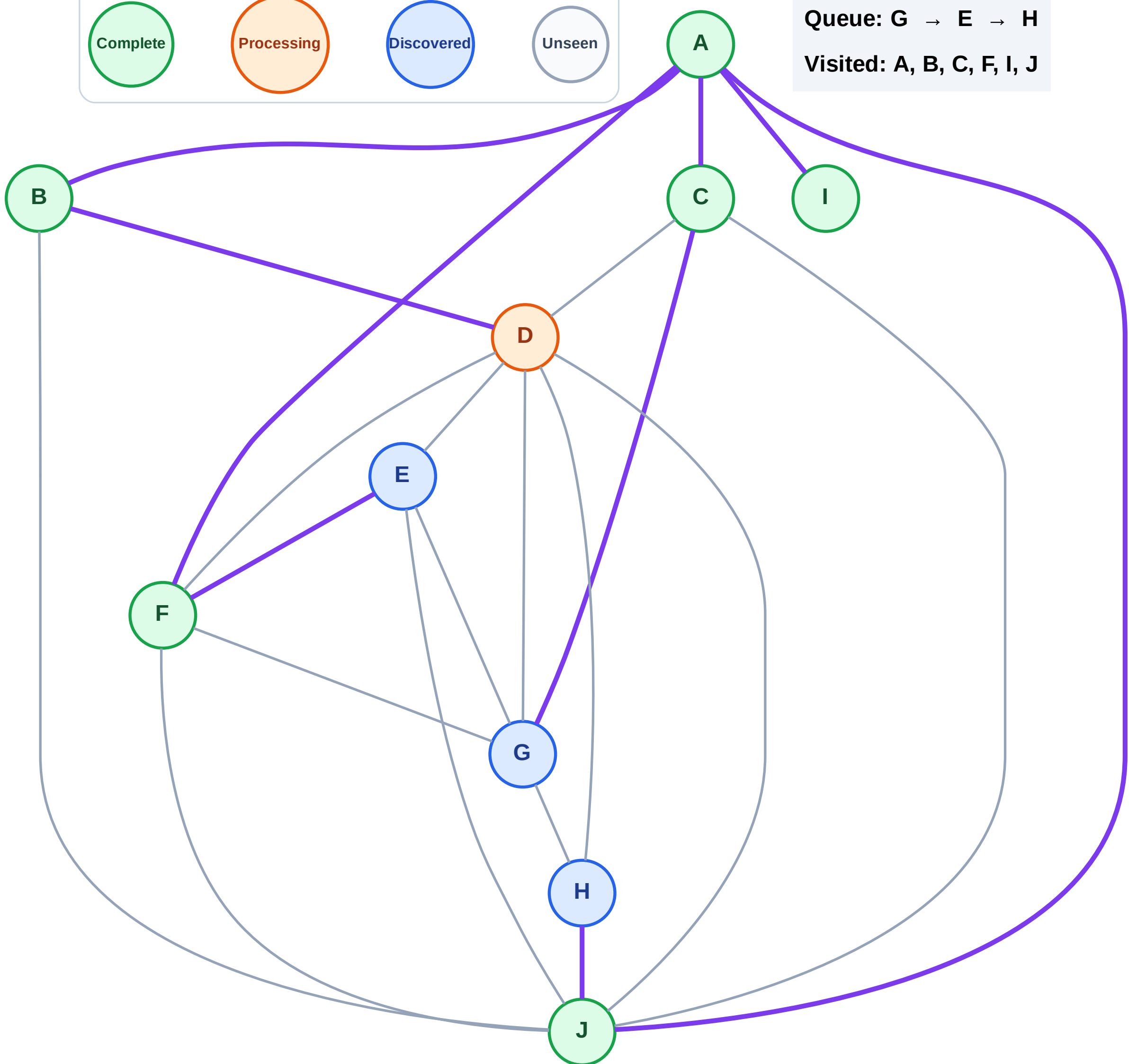


BFS Traversal

Step 14: Process node D



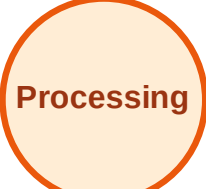
Queue: G → E → H
Visited: A, B, C, F, I, J



BFS Traversal

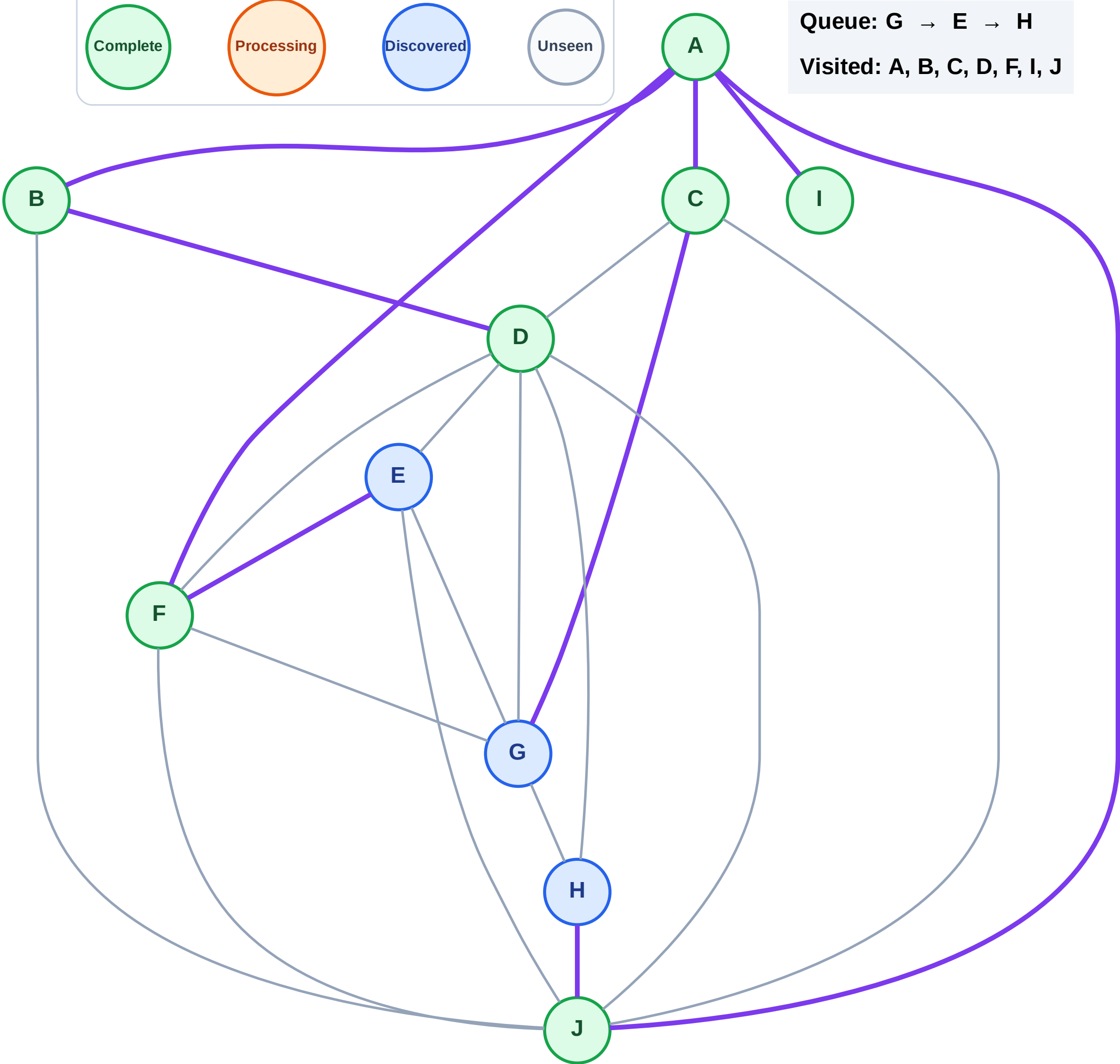
Step 15: No new neighbors from D

Legend



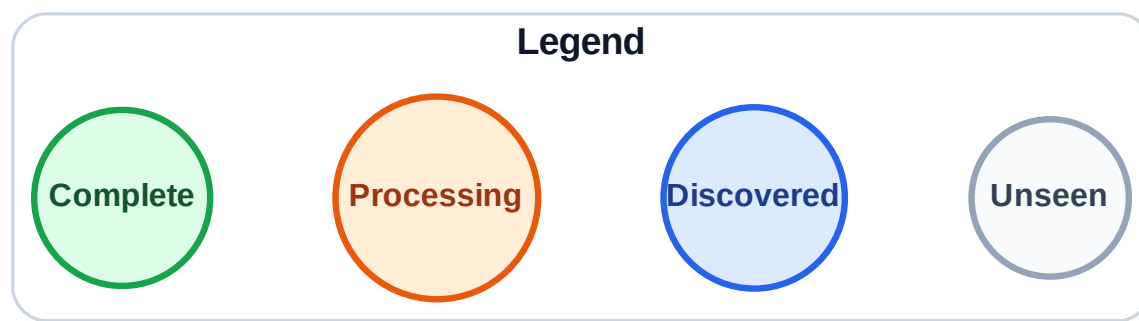
Queue: G → E → H

Visited: A, B, C, D, F, I, J

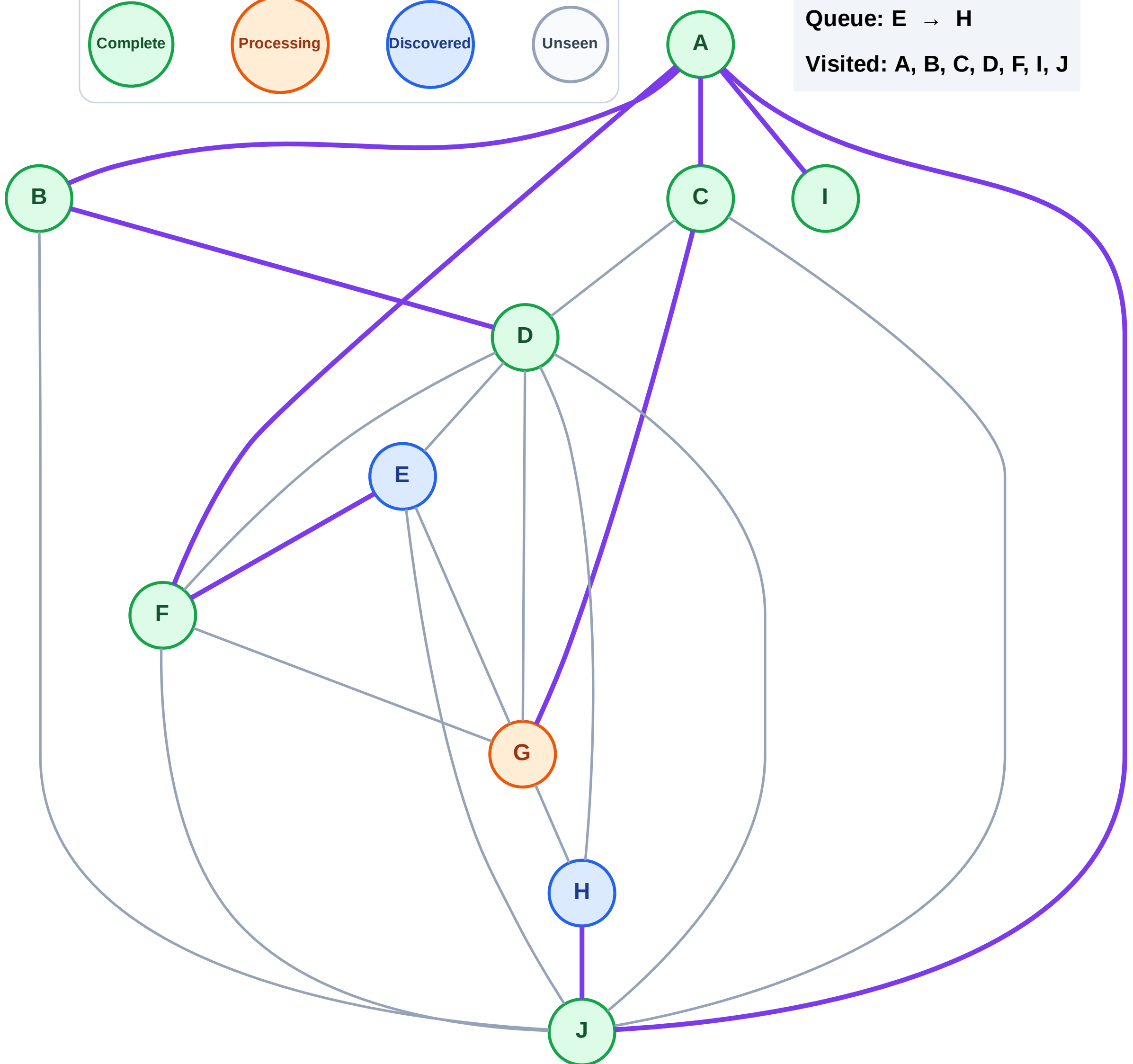


BFS Traversal

Step 16: Process node G



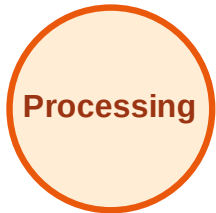
Queue: E → H
Visited: A, B, C, D, F, I, J



BFS Traversal

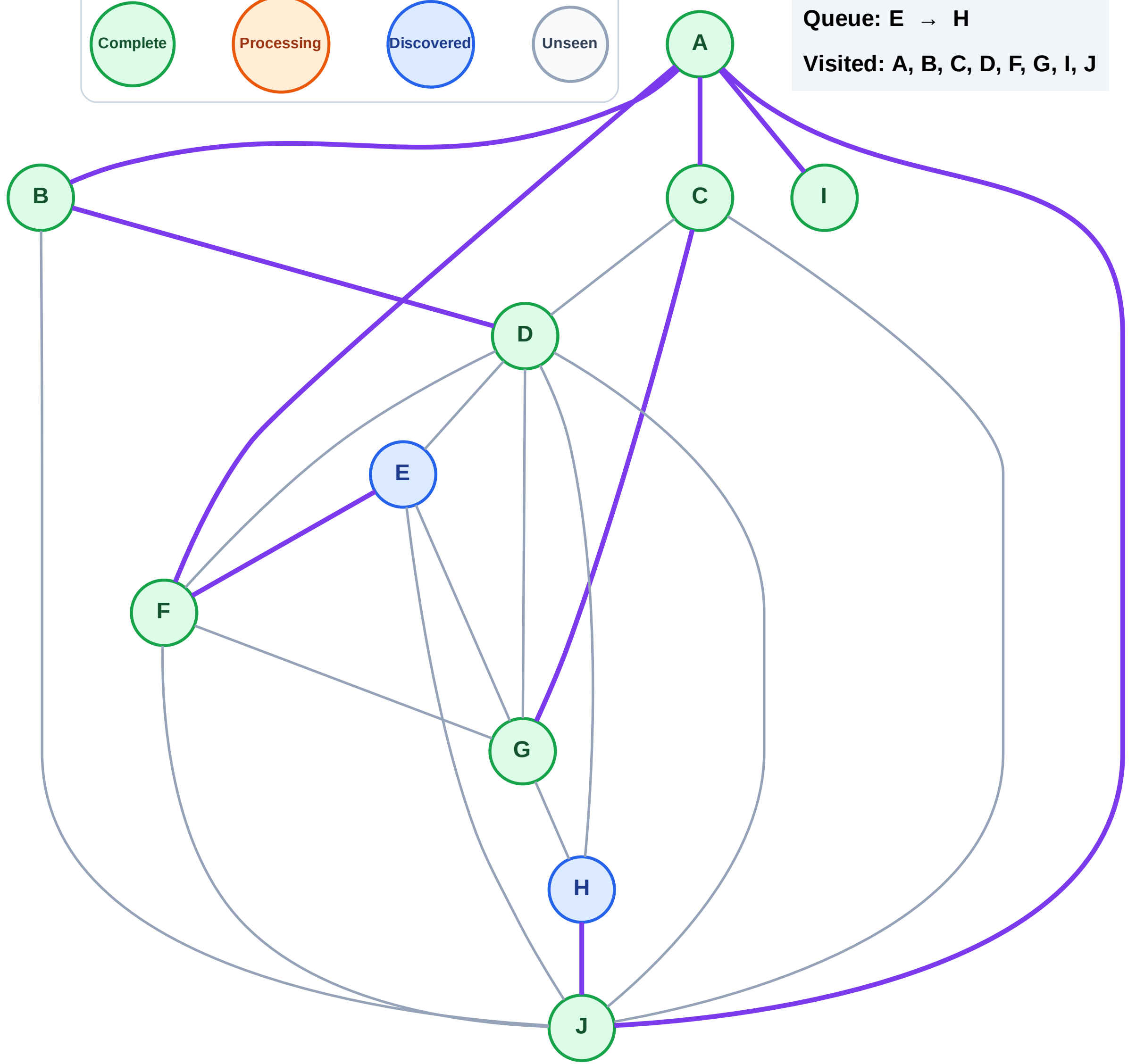
Step 17: No new neighbors from G

Legend



Queue: E → H

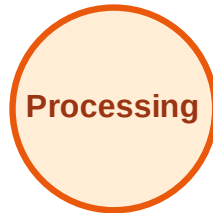
Visited: A, B, C, D, F, G, I, J



BFS Traversal

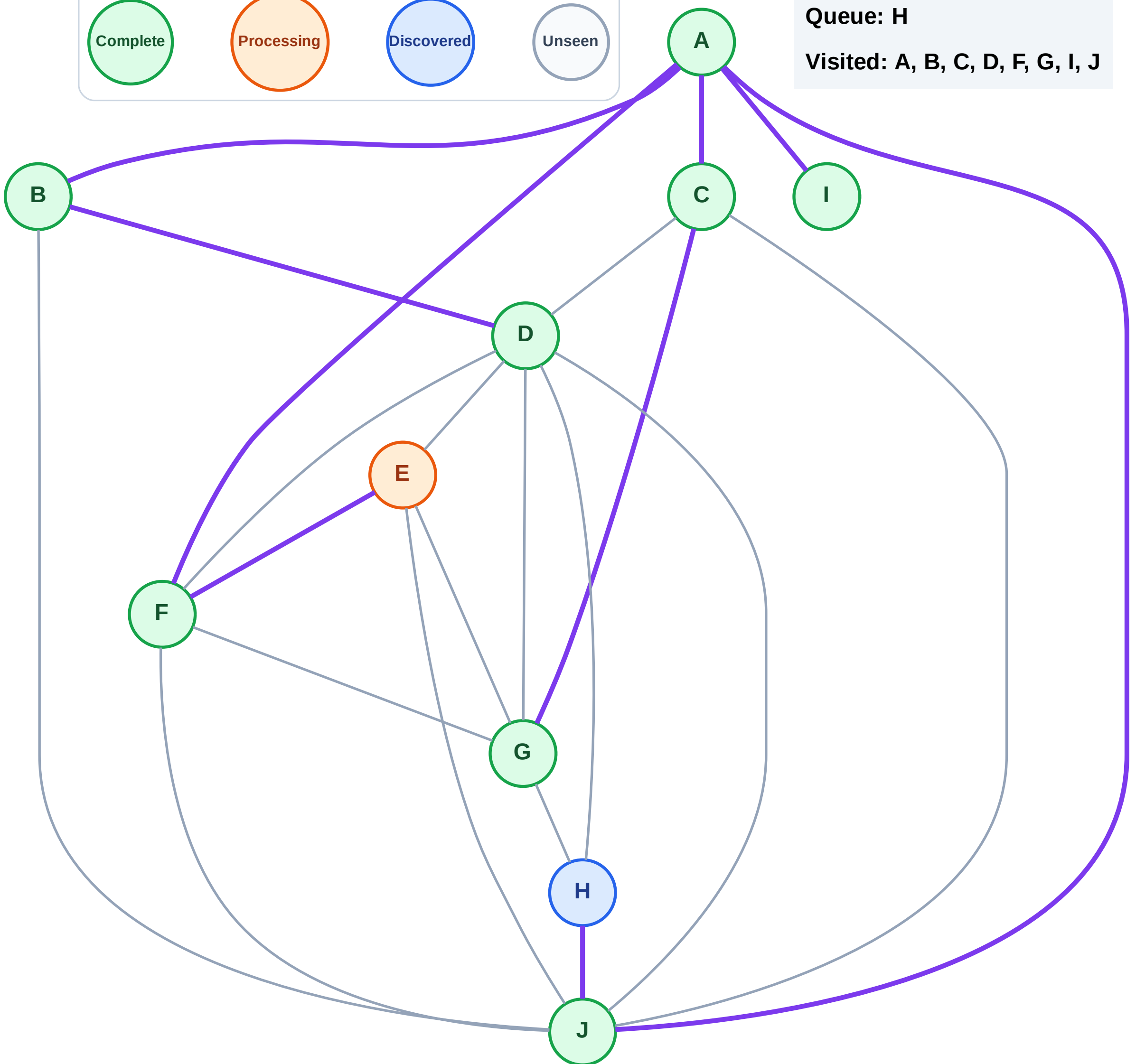
Step 18: Process node E

Legend



Queue: H

Visited: A, B, C, D, F, G, I, J



BFS Traversal

Step 19: No new neighbors from E

Legend

Complete

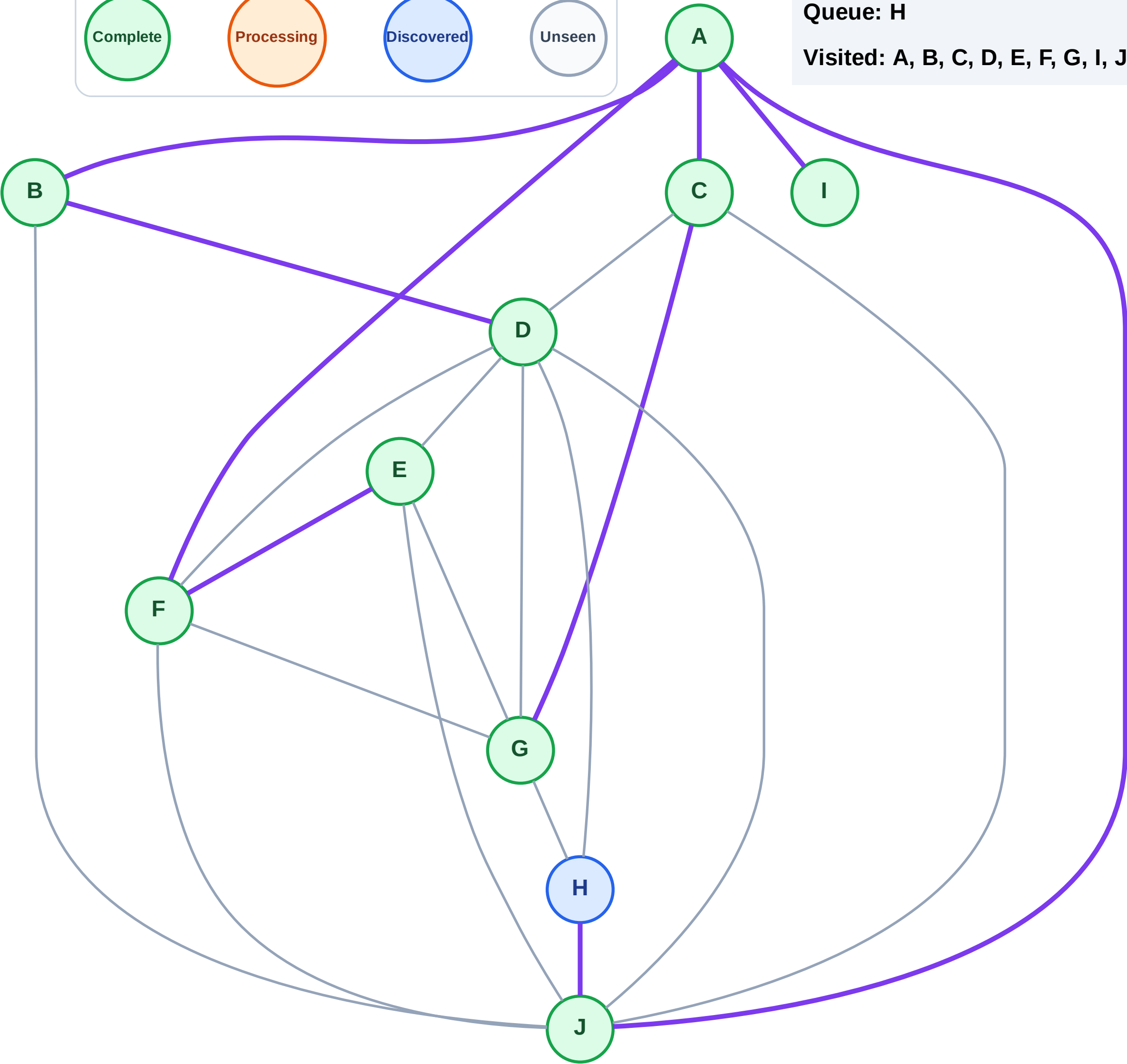
Processing

Discovered

Unseen

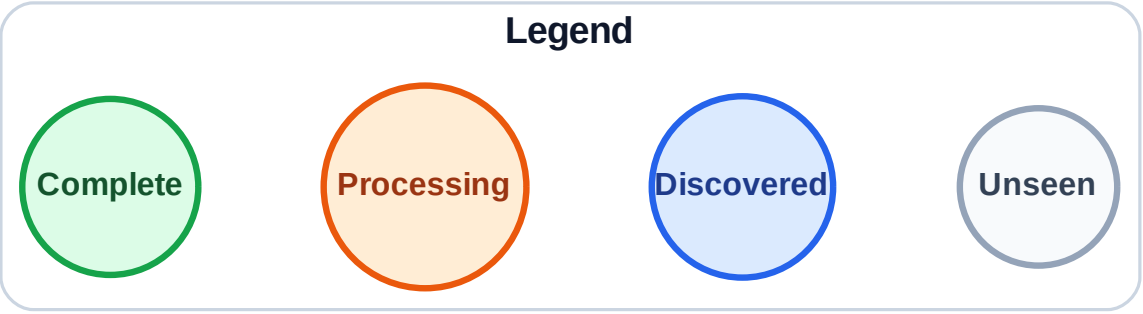
Queue: H

Visited: A, B, C, D, E, F, G, I, J



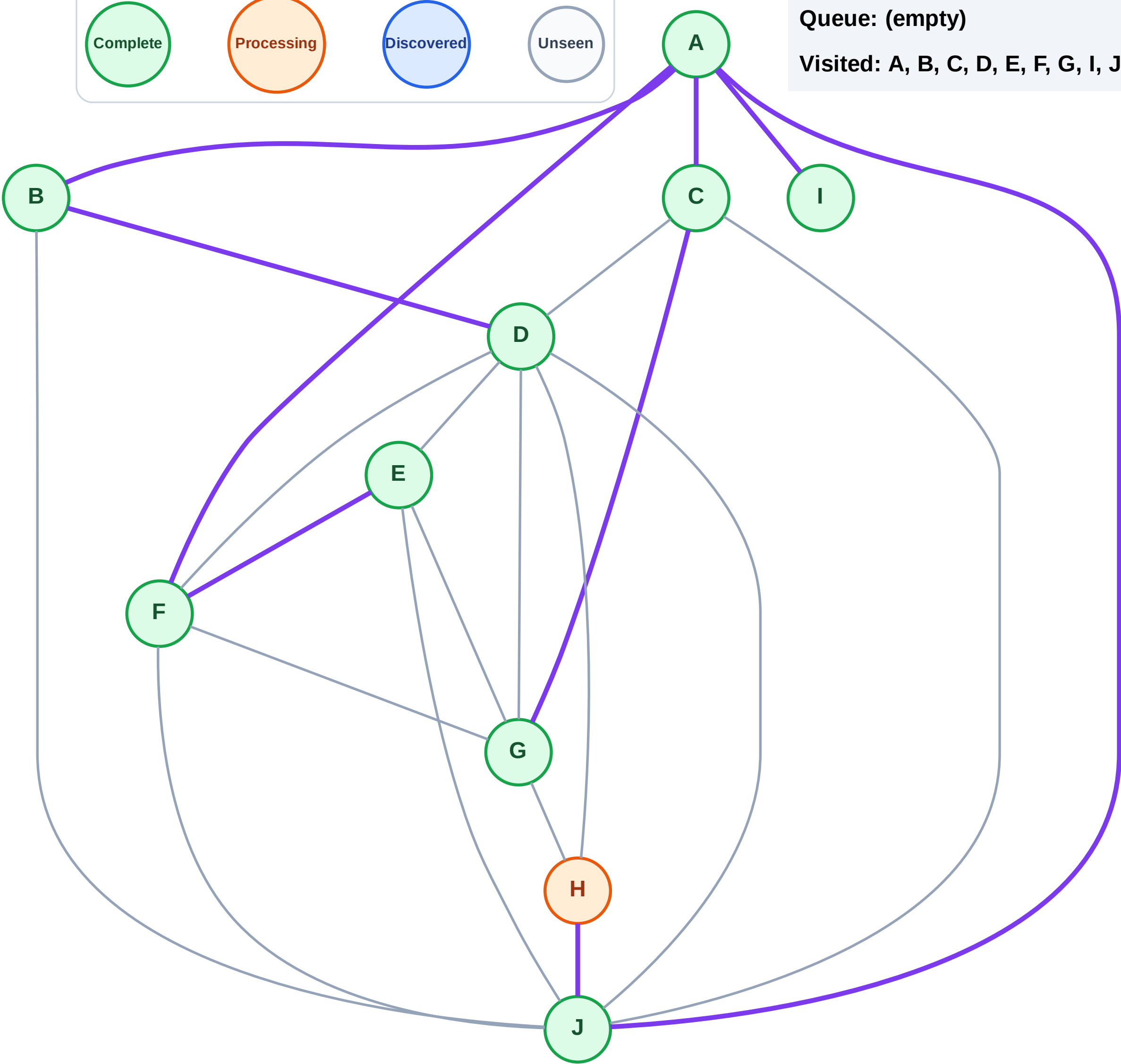
BFS Traversal

Step 20: Process node H



Queue: (empty)

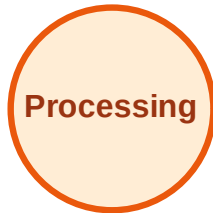
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

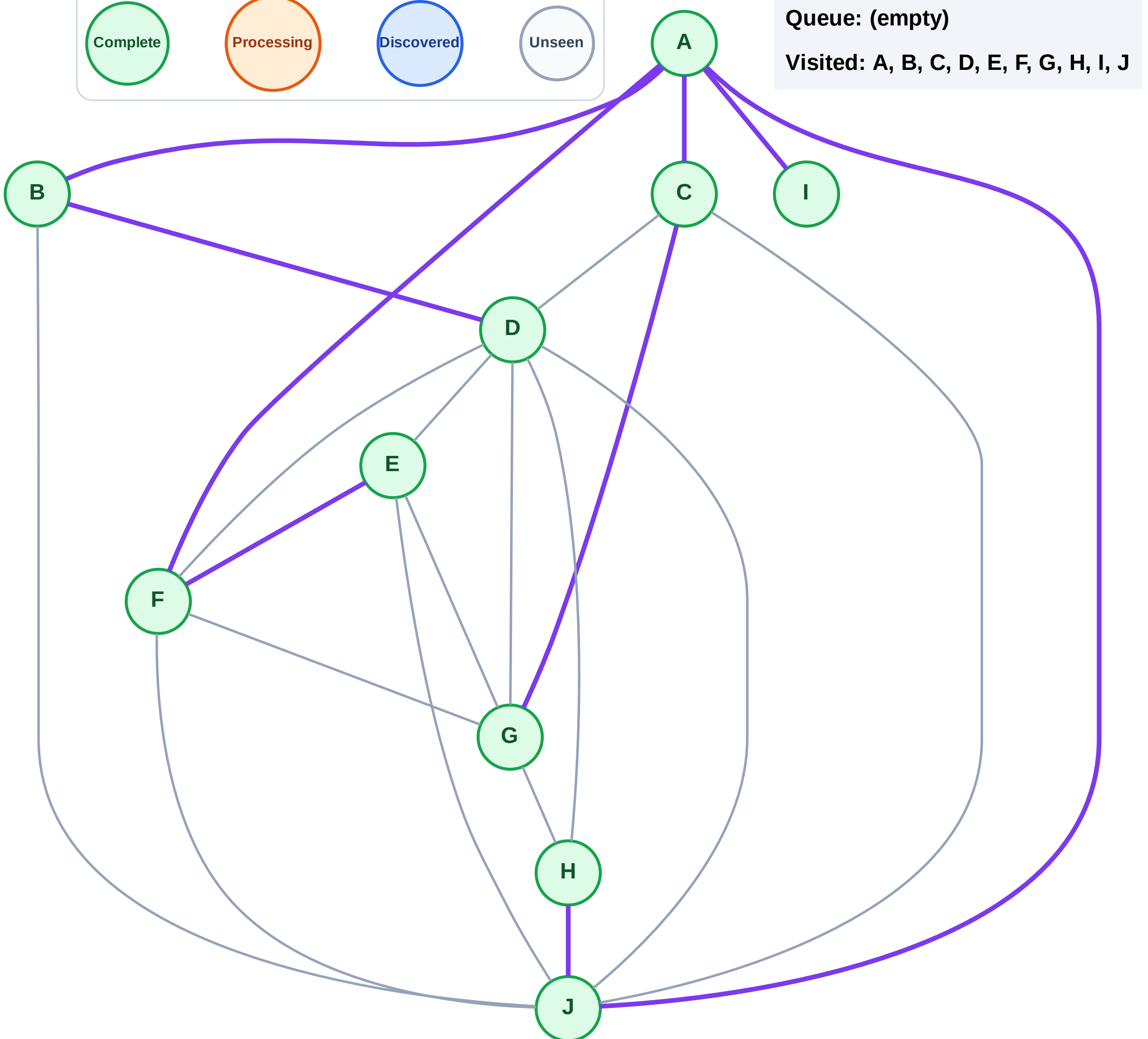
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

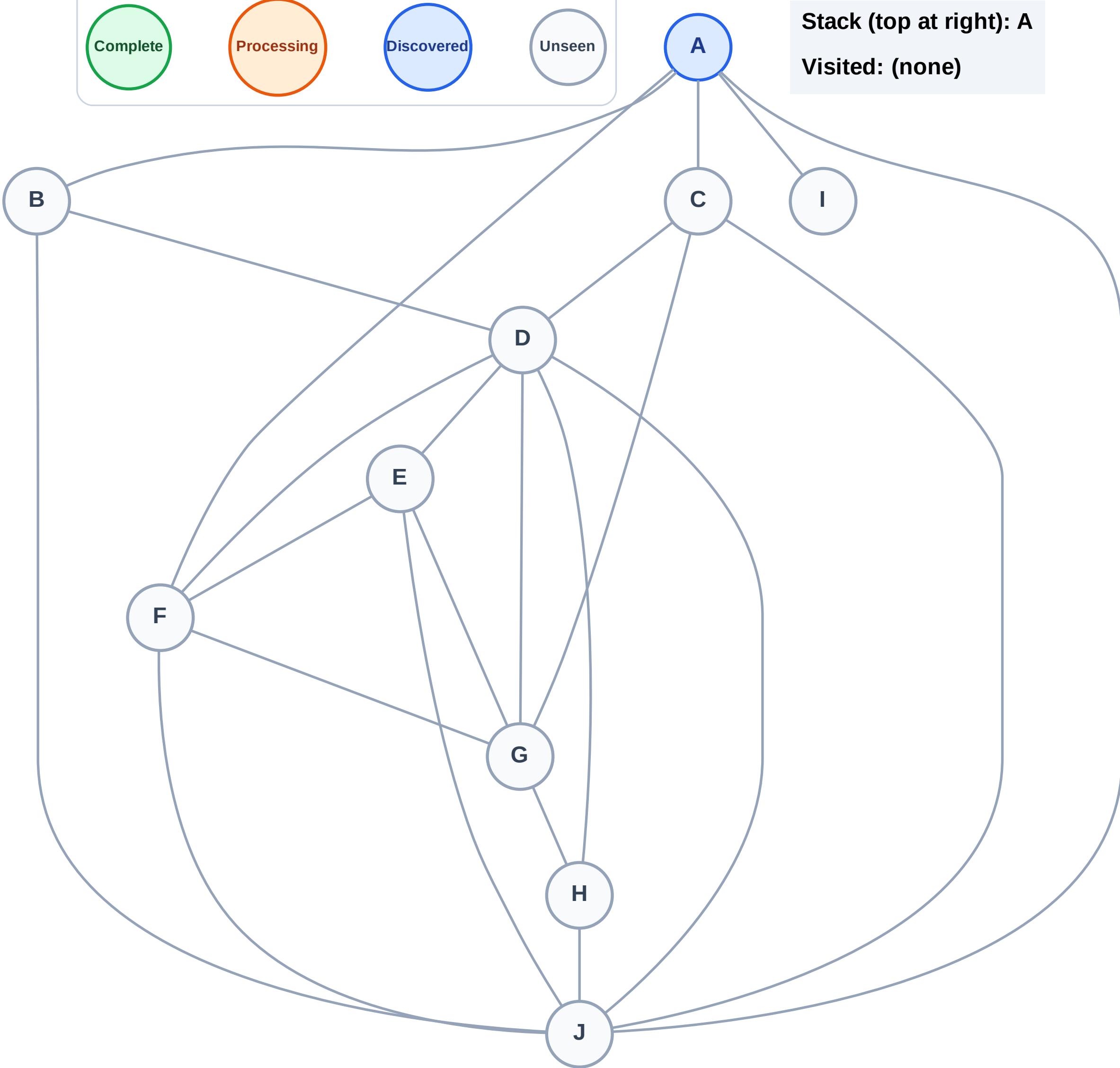
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

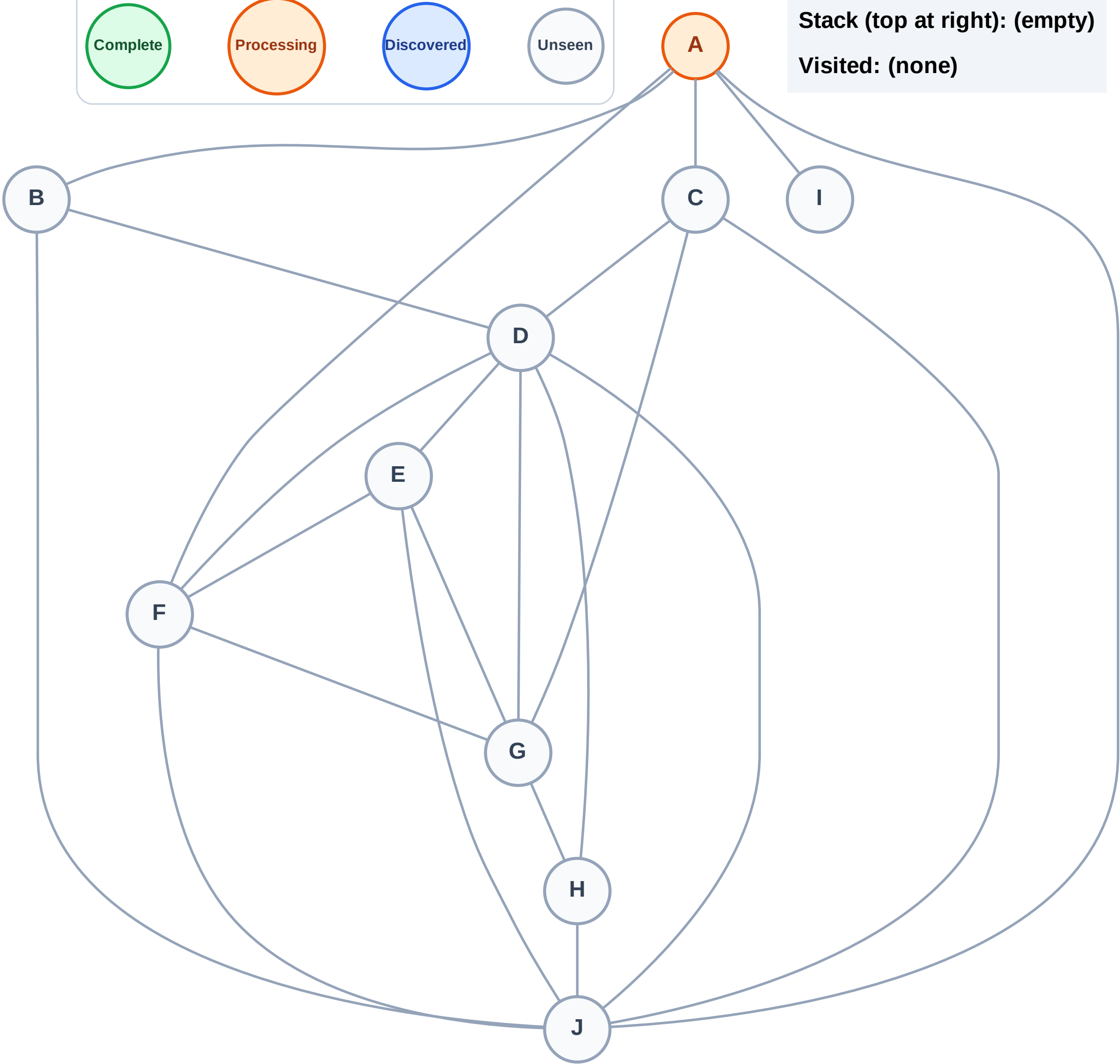
Complete

Processing

Discovered

Unseen

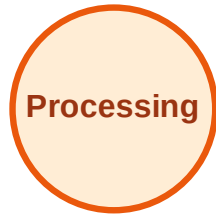
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

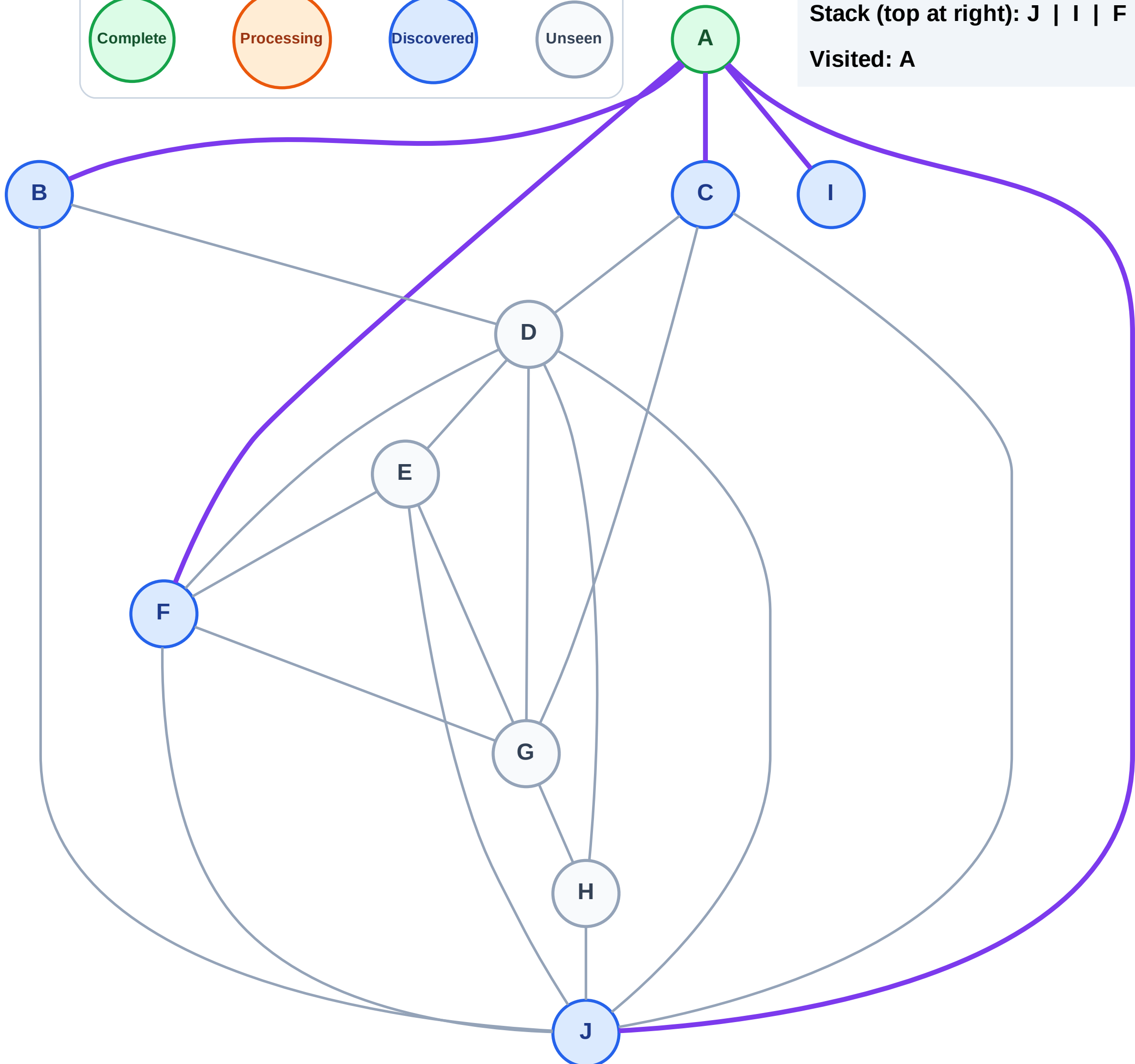
Step 3: Discover J, I, F, C, B from A

Legend



Stack (top at right): J | I | F | C | B

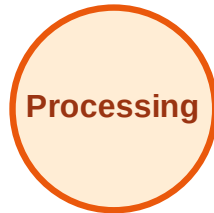
Visited: A



DFS Traversal

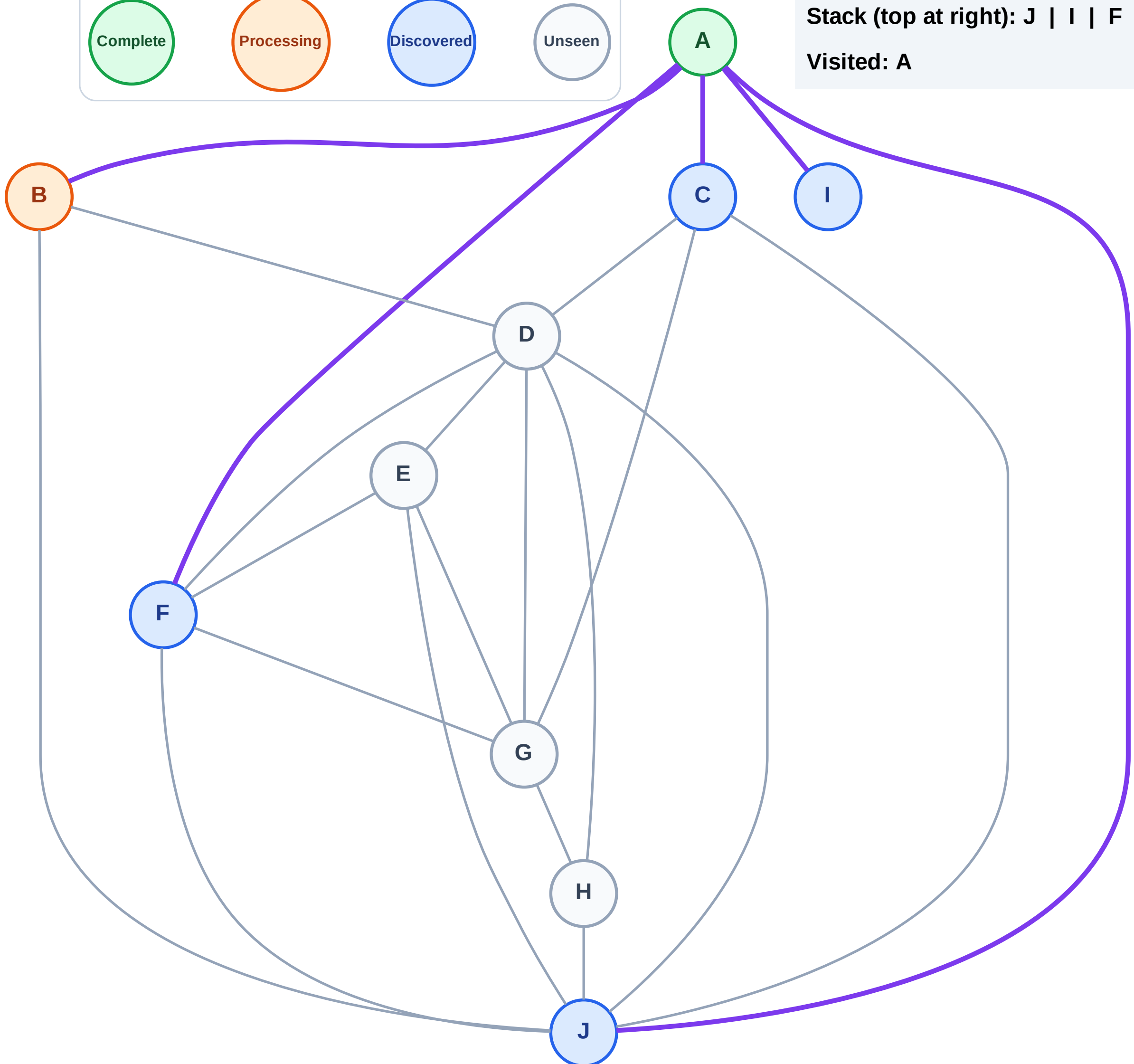
Step 4: Process node B

Legend



Stack (top at right): J | I | F | C

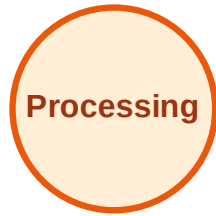
Visited: A



DFS Traversal

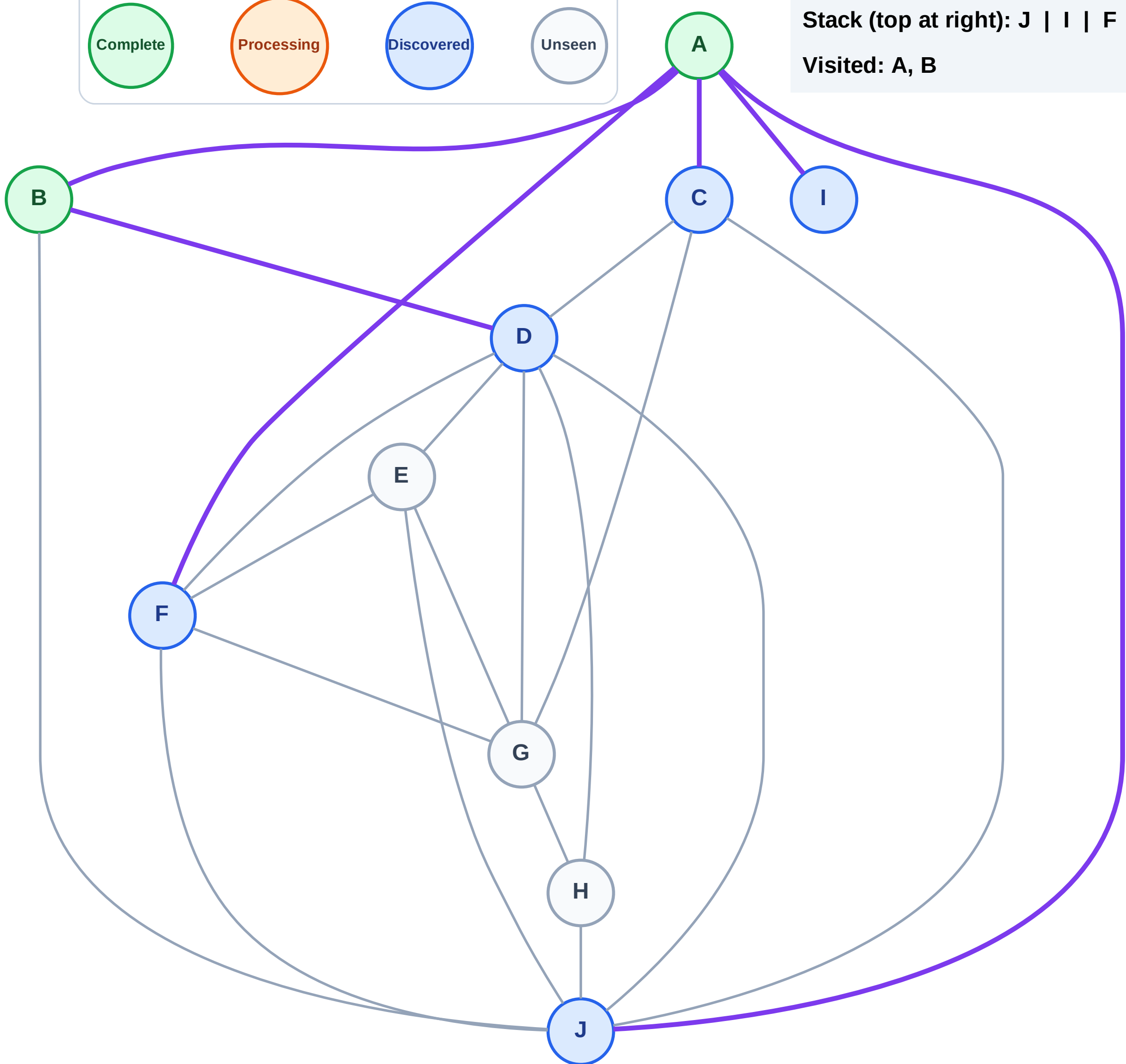
Step 5: Discover D from B

Legend



Stack (top at right): J | I | F | C | D

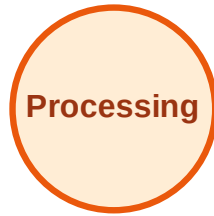
Visited: A, B



DFS Traversal

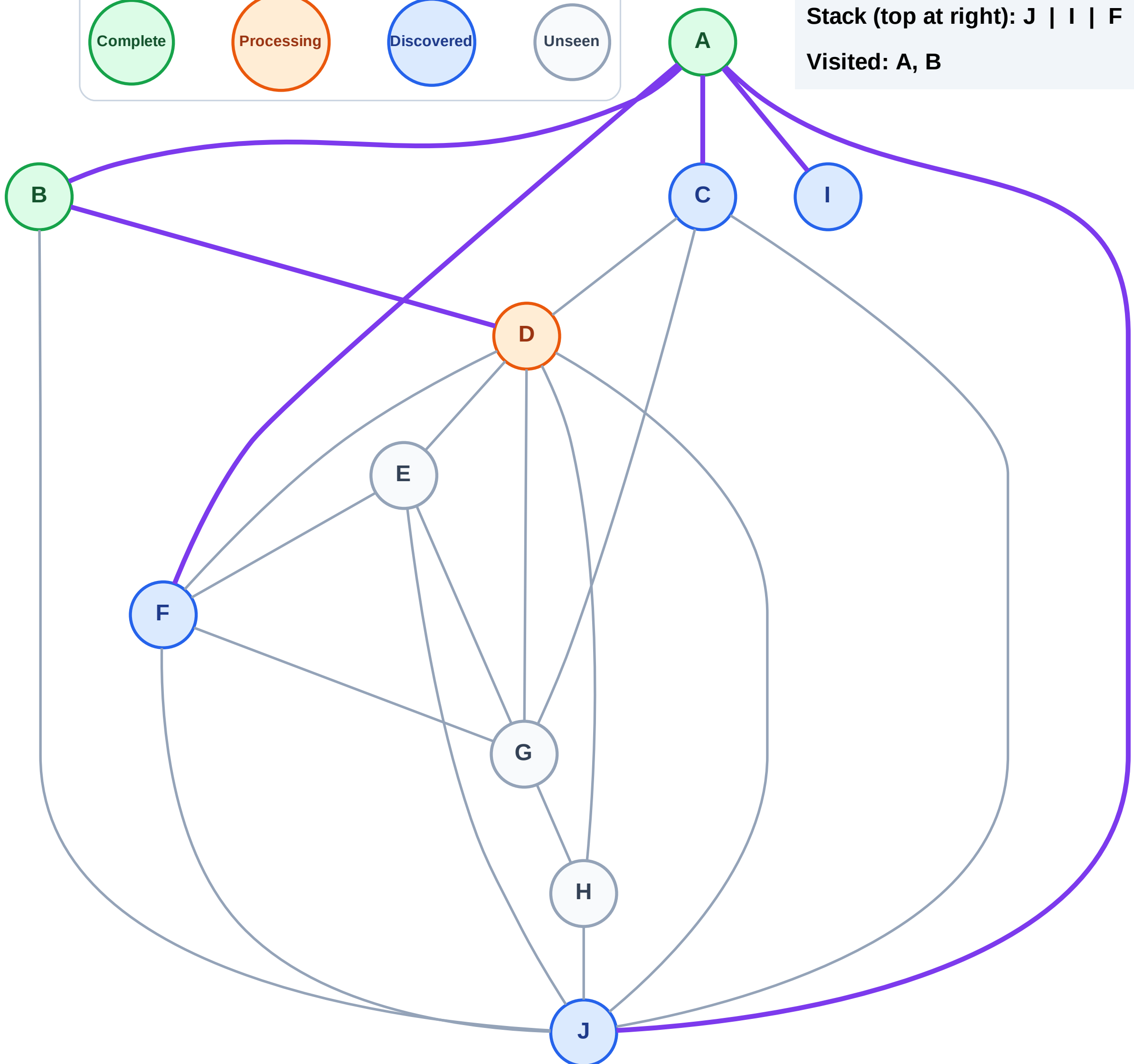
Step 6: Process node D

Legend



Stack (top at right): J | I | F | C

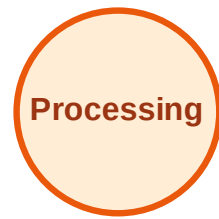
Visited: A, B



DFS Traversal

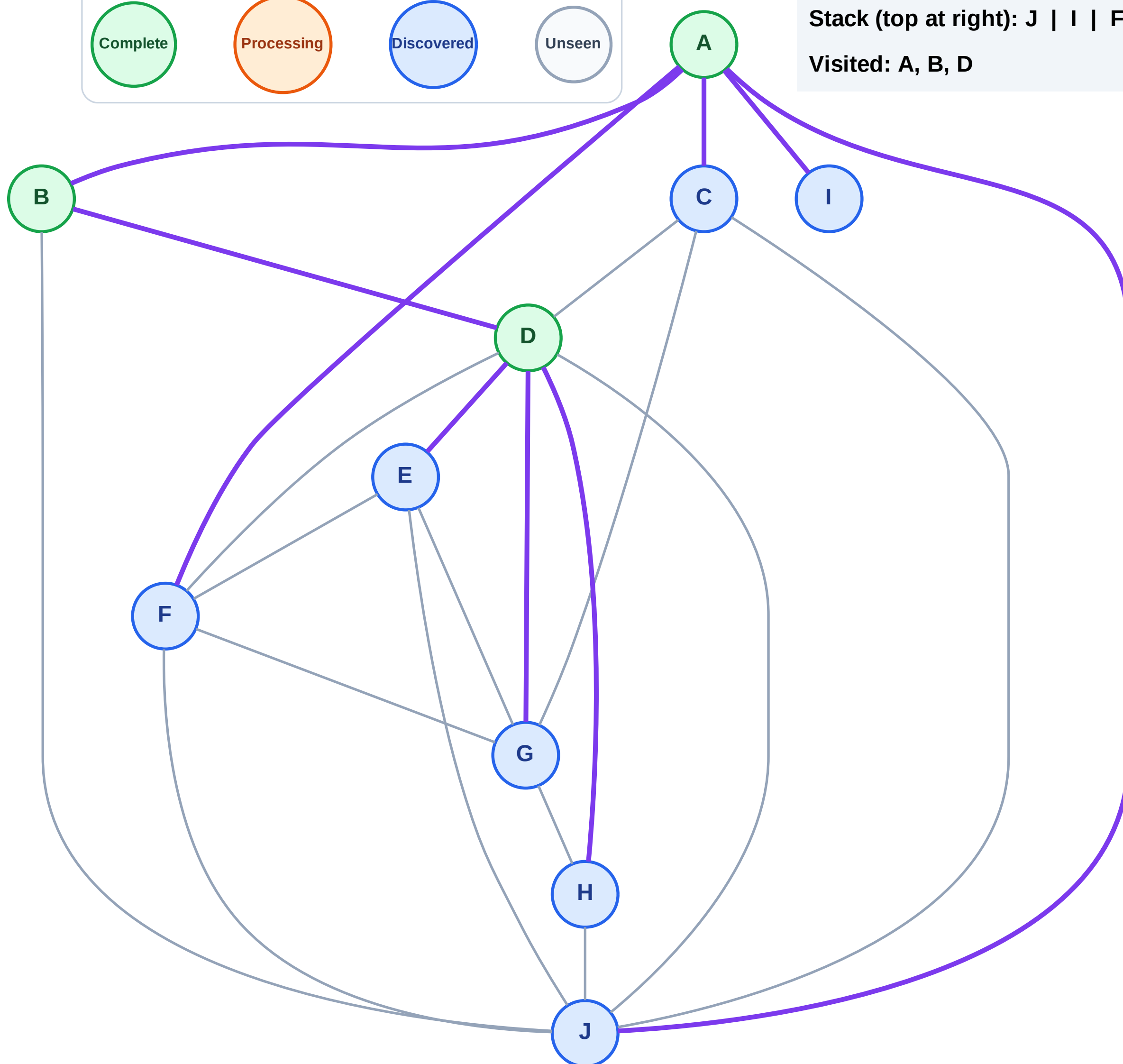
Step 7: Discover H, G, E from D

Legend



Stack (top at right): J | I | F | C | H | G | E

Visited: A, B, D



DFS Traversal

Step 8: Process node E

Legend

Complete

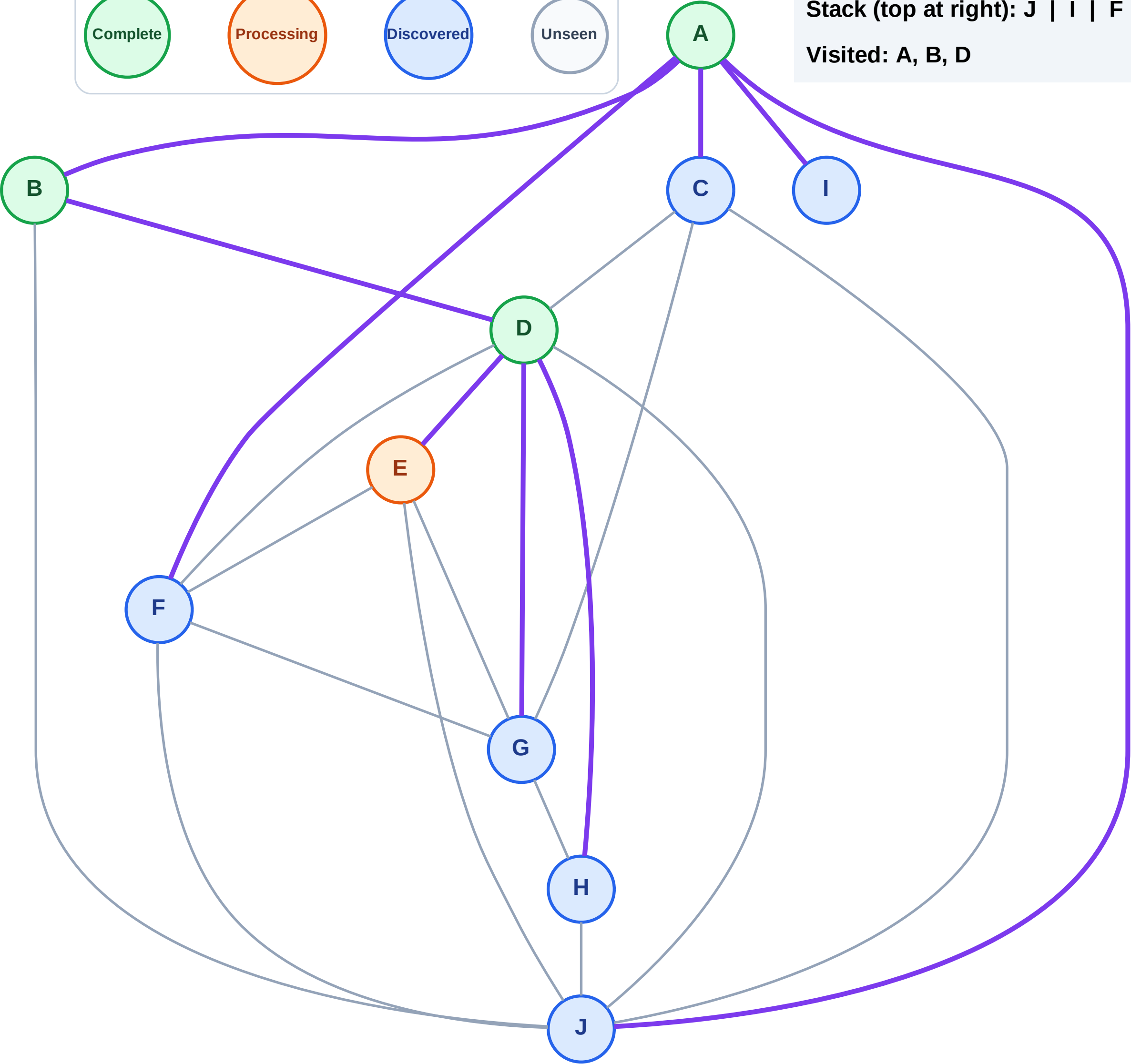
Processing

Discovered

Unseen

Stack (top at right): J | I | F | C | H | G

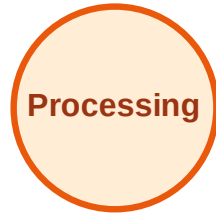
Visited: A, B, D



DFS Traversal

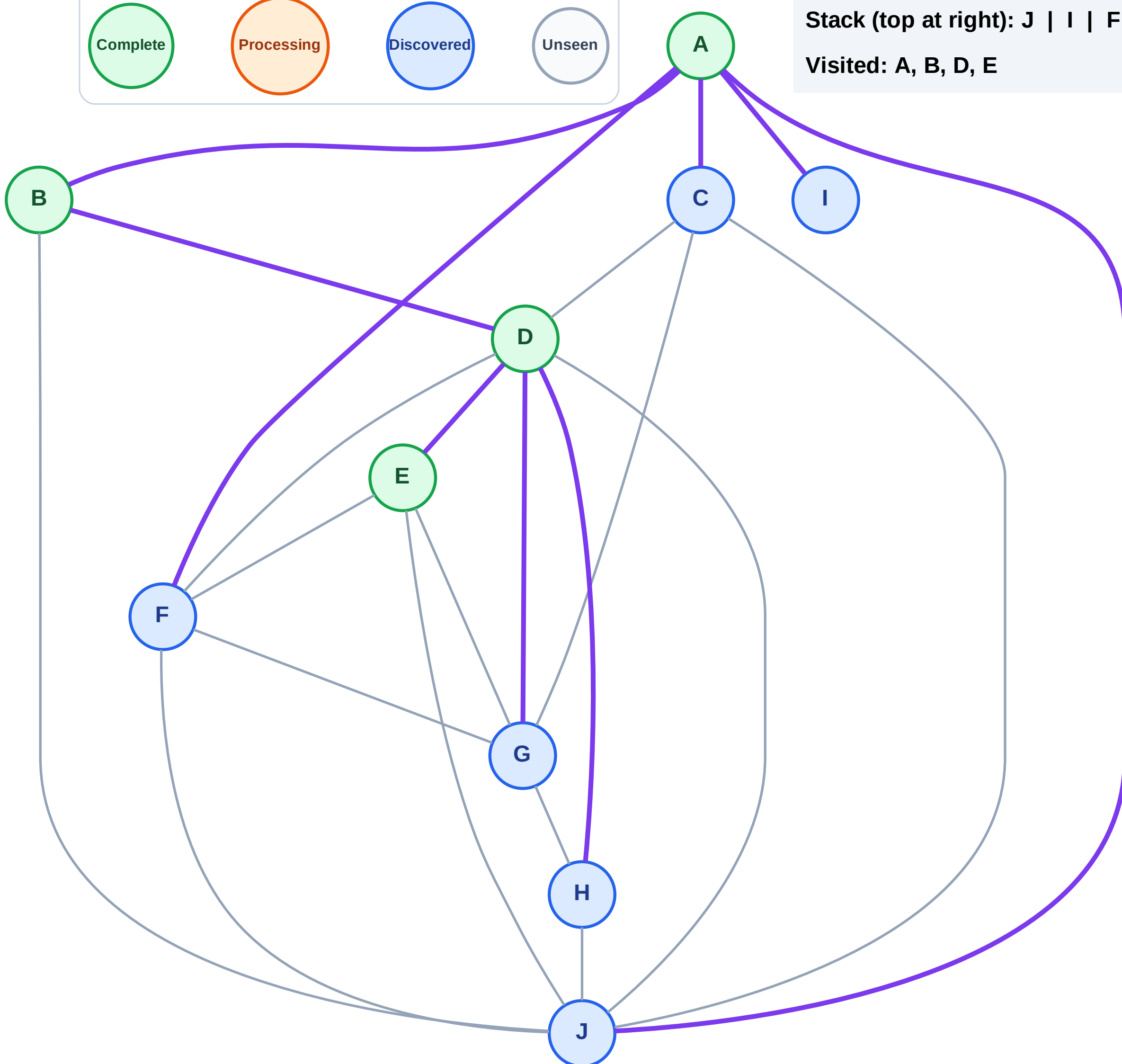
Step 9: No new neighbors from E

Legend



Stack (top at right): J | I | F | C | H | G

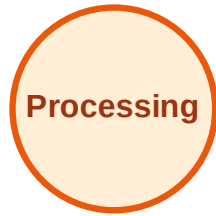
Visited: A, B, D, E



DFS Traversal

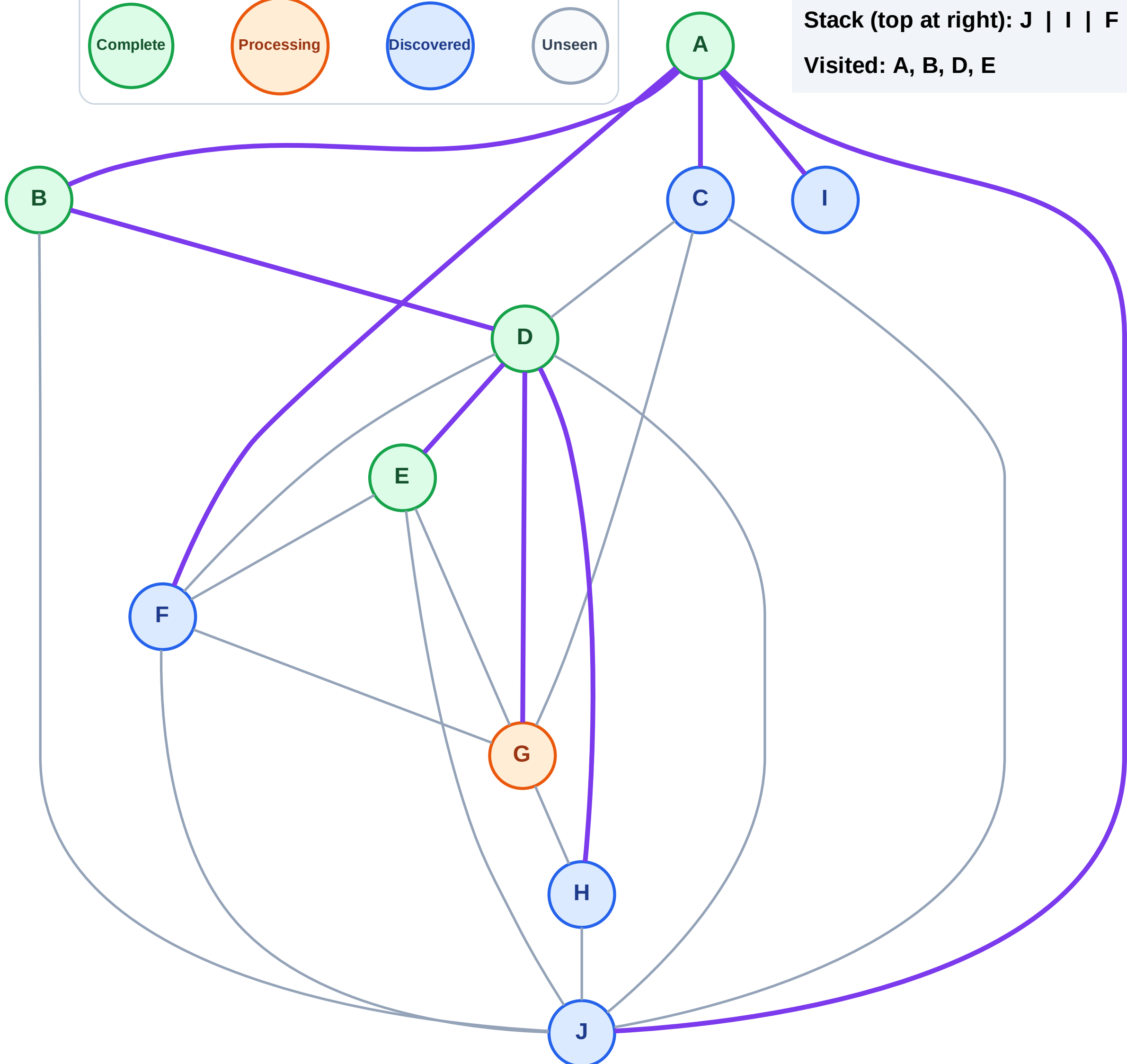
Step 10: Process node G

Legend



Stack (top at right): J | I | F | C | H

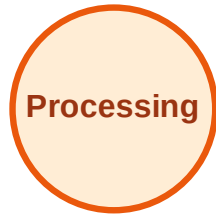
Visited: A, B, D, E



DFS Traversal

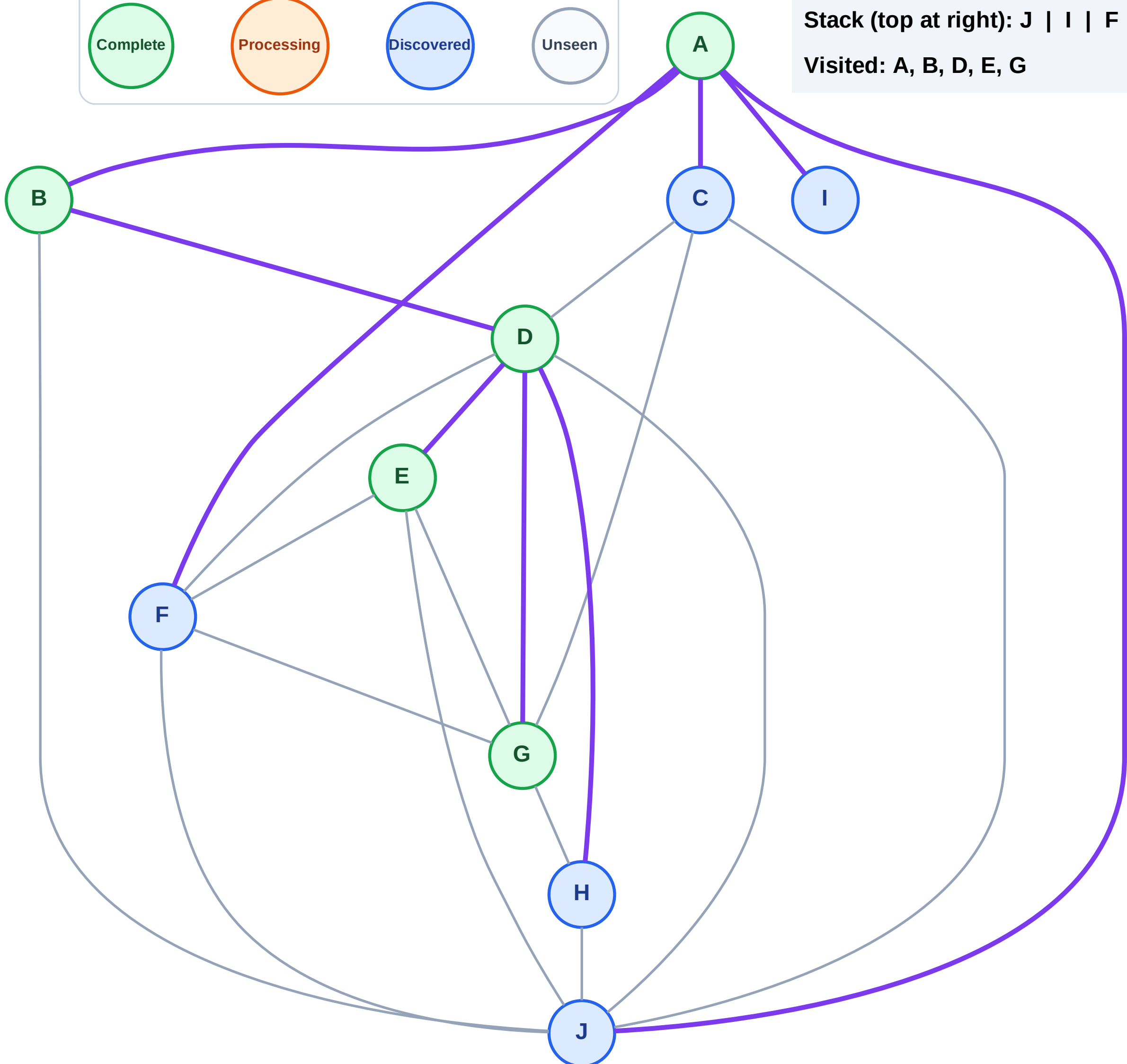
Step 11: No new neighbors from G

Legend



Stack (top at right): J | I | F | C | H

Visited: A, B, D, E, G



DFS Traversal

Step 12: Process node H

Legend

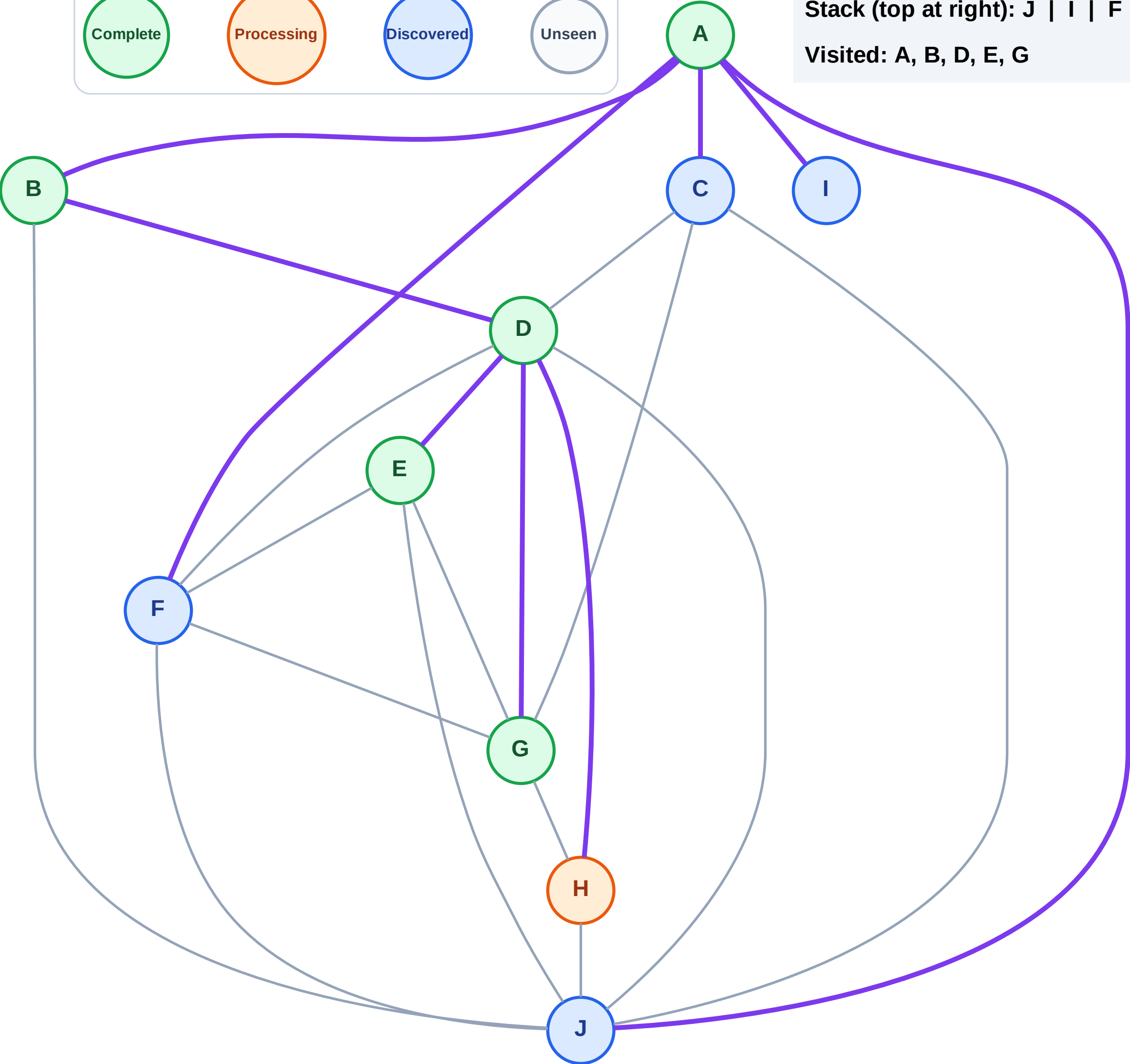
Complete

Processing

Discovered

Unseen

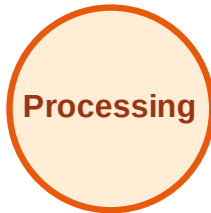
Stack (top at right): J | I | F | C
Visited: A, B, D, E, G



DFS Traversal

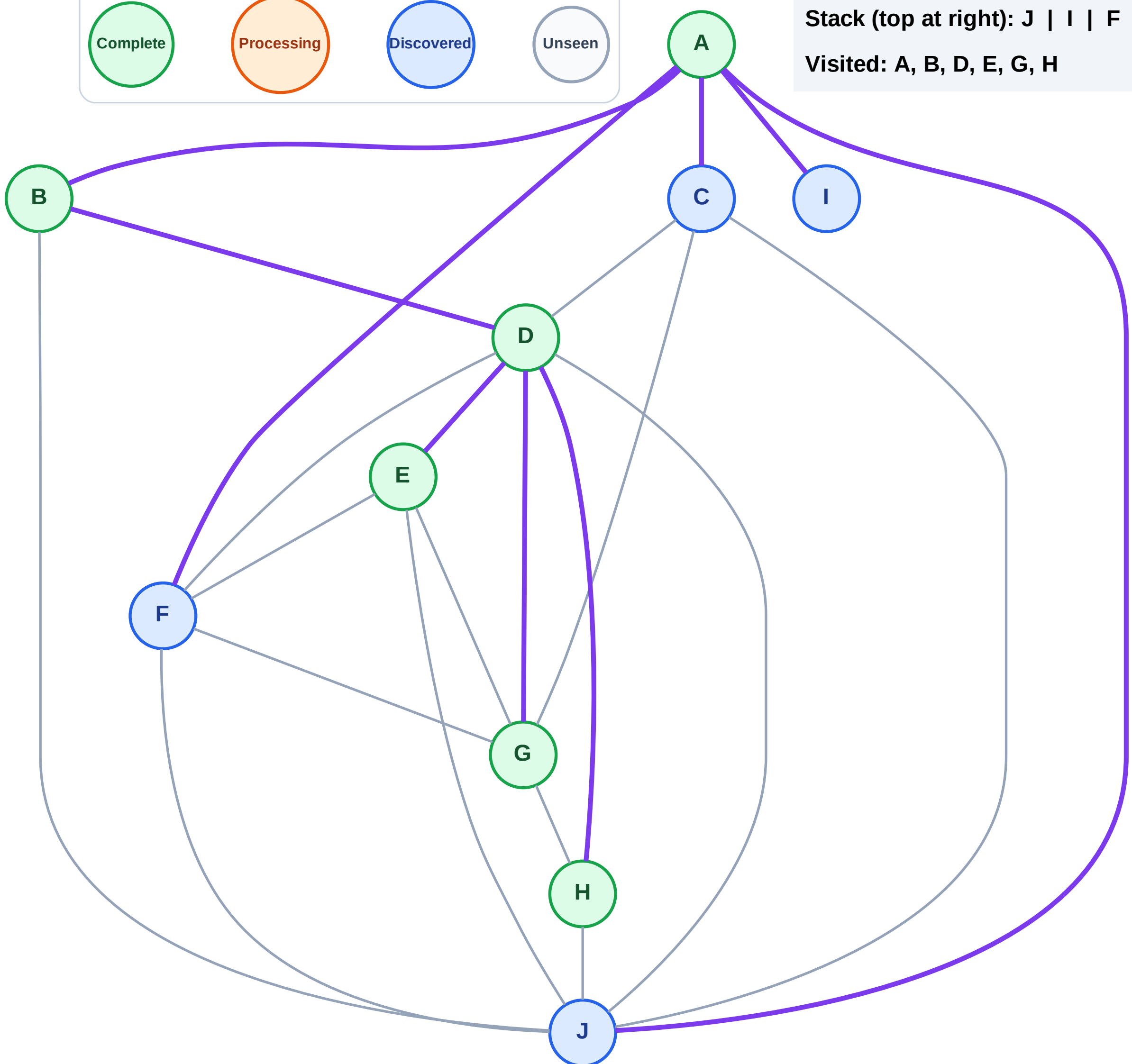
Step 13: No new neighbors from H

Legend



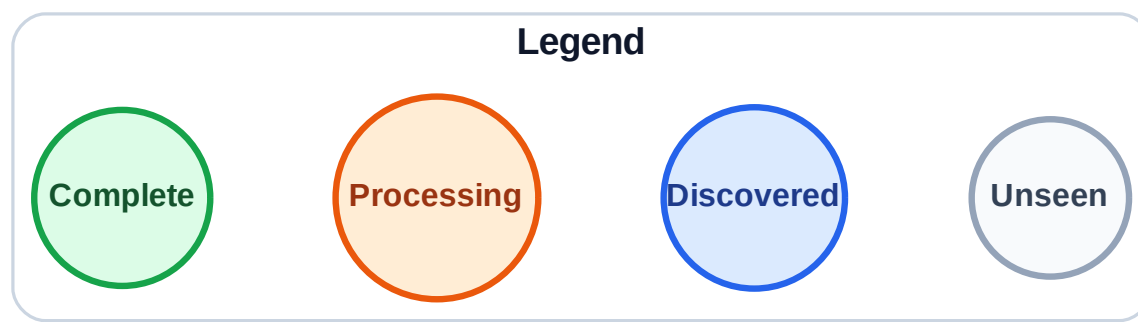
Stack (top at right): J | I | F | C

Visited: A, B, D, E, G, H

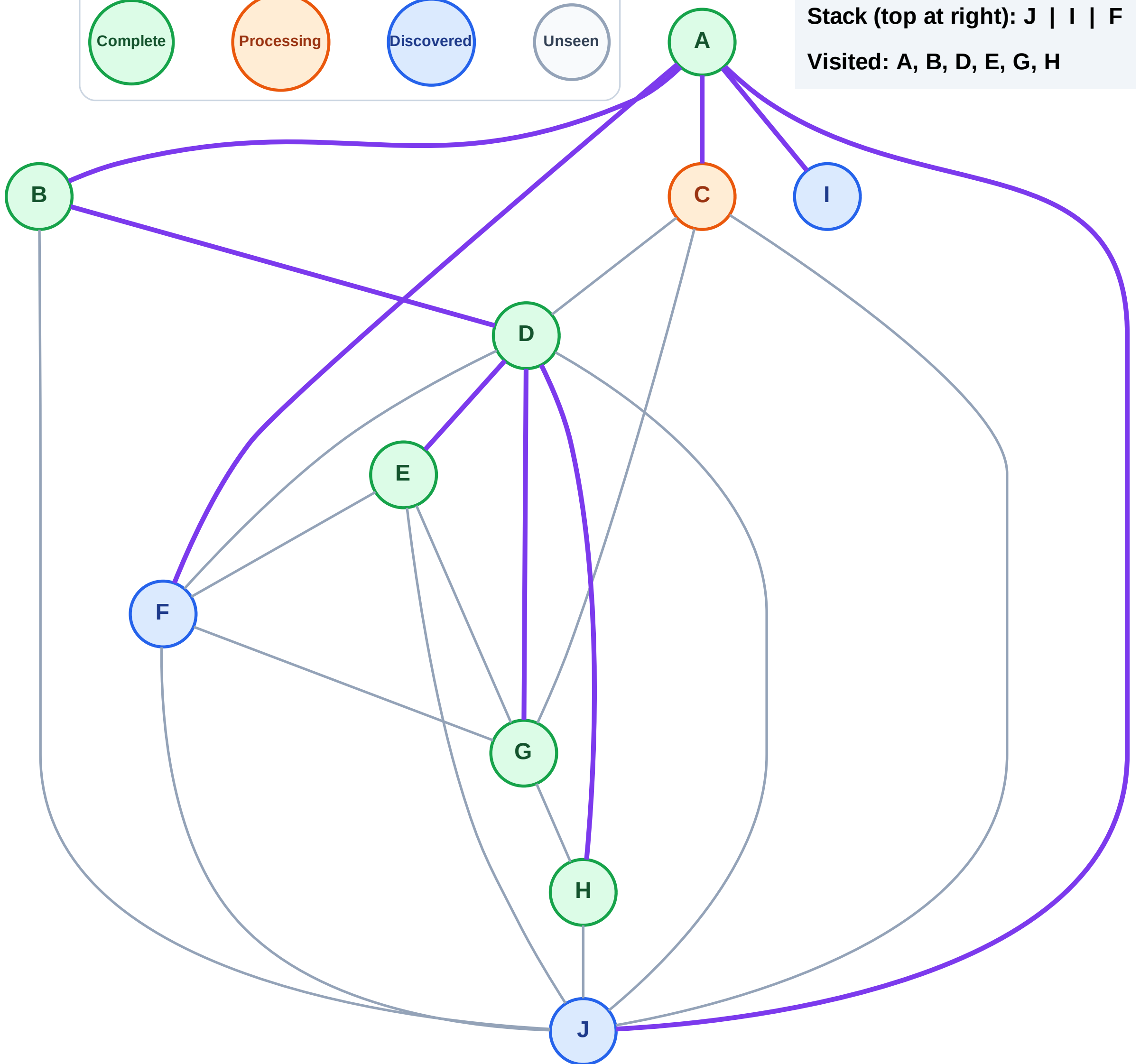


DFS Traversal

Step 14: Process node C



Stack (top at right): J | I | F
Visited: A, B, D, E, G, H



DFS Traversal

Step 15: No new neighbors from C

Legend

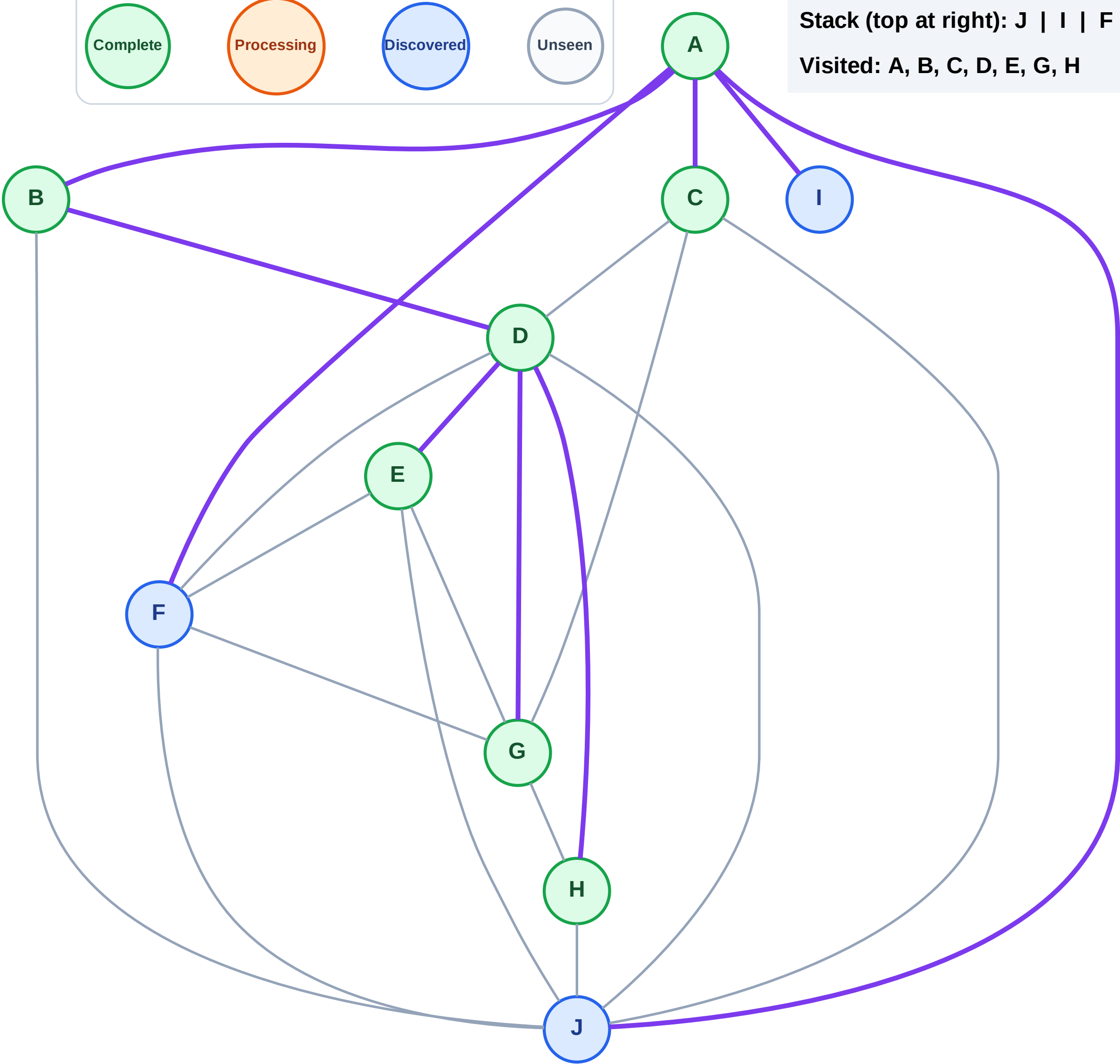
Complete

Processing

Discovered

Unseen

Stack (top at right): J | I | F
Visited: A, B, C, D, E, G, H



DFS Traversal

Step 16: Process node F

Legend

Complete

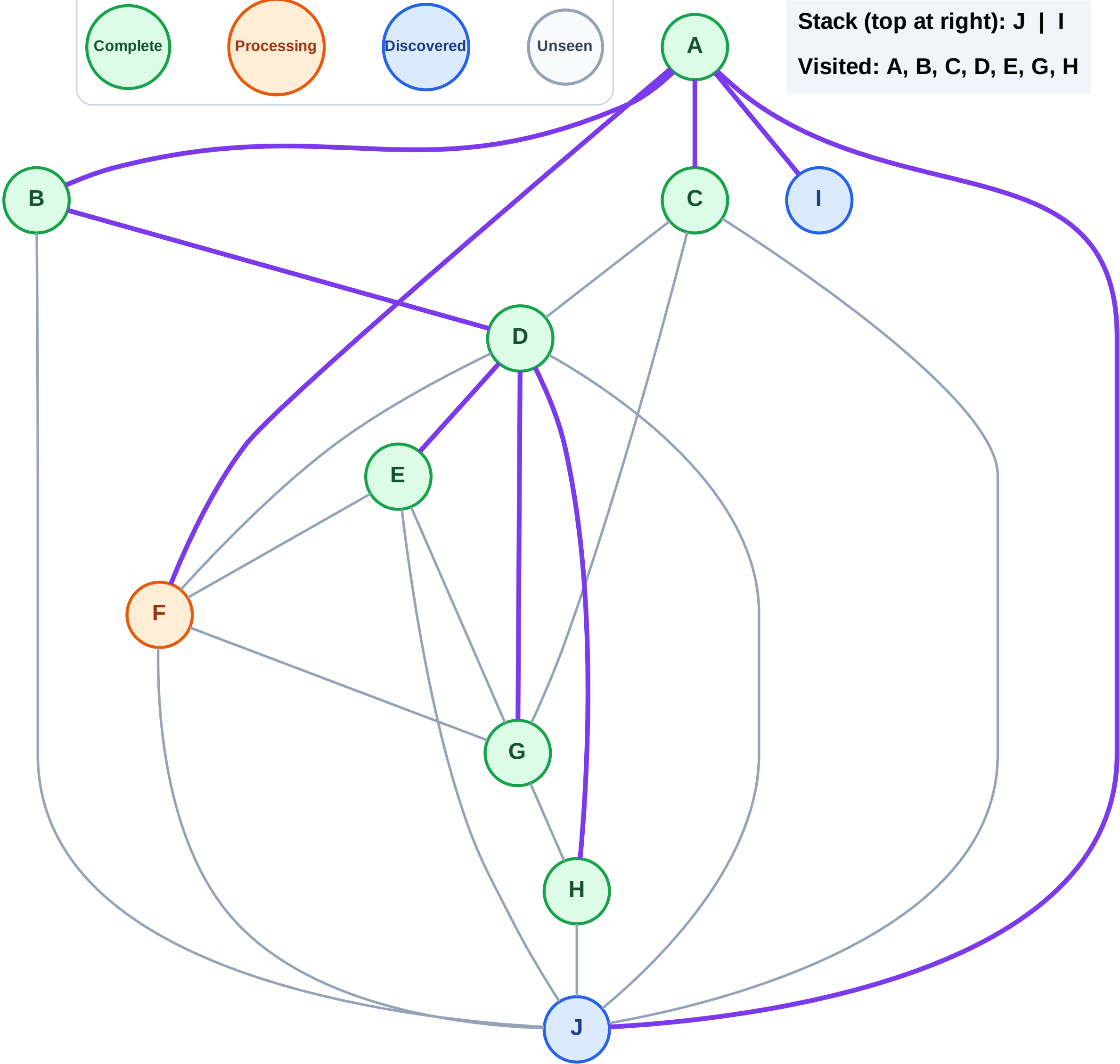
Processing

Discovered

Unseen

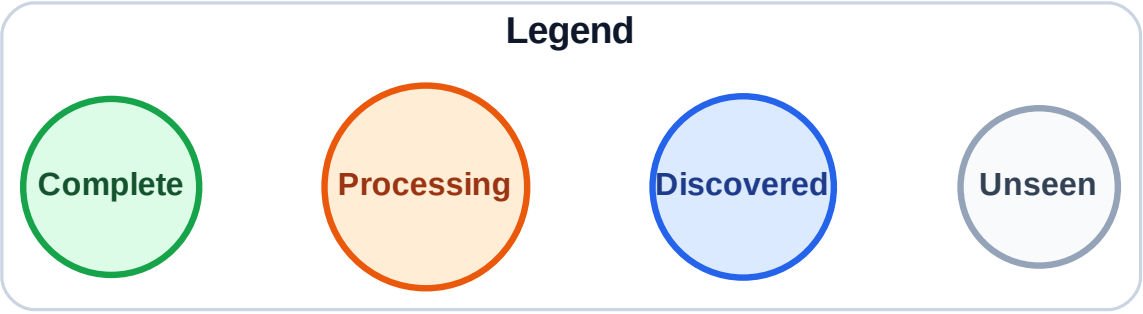
Stack (top at right): J | I

Visited: A, B, C, D, E, G, H

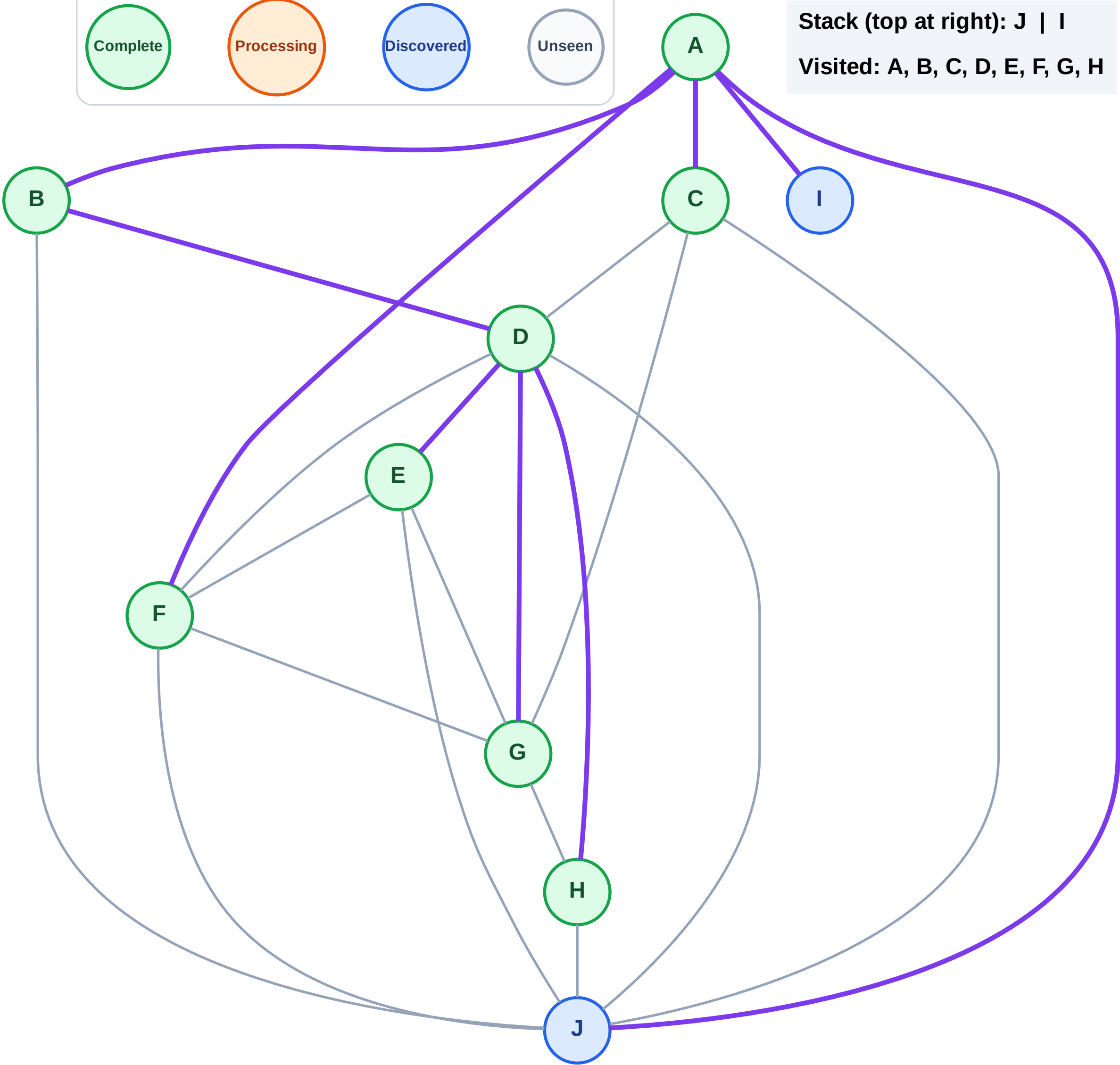


DFS Traversal

Step 17: No new neighbors from F



Stack (top at right): J | I
Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 18: Process node I

Legend

Complete

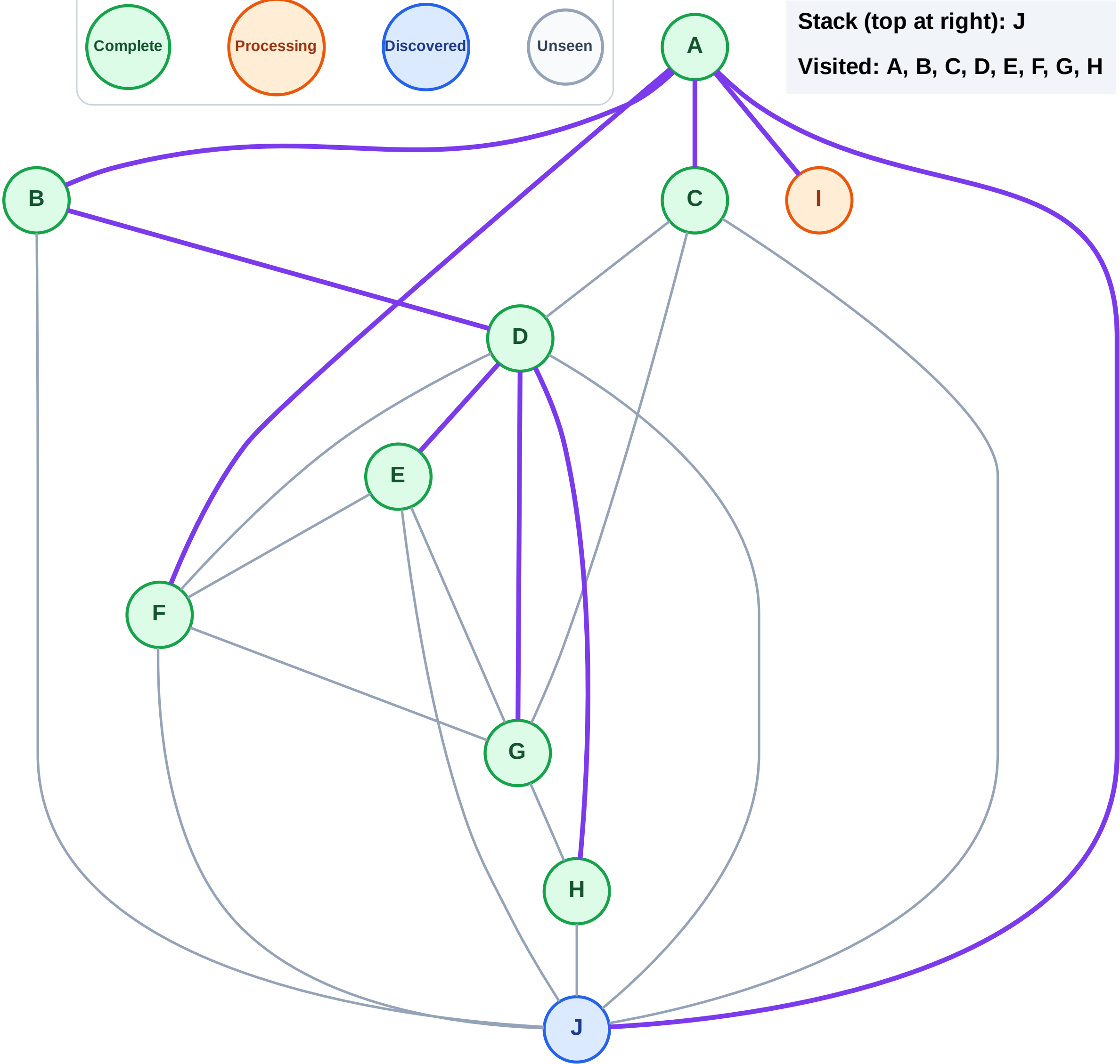
Processing

Discovered

Unseen

Stack (top at right): J

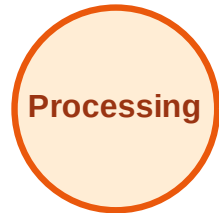
Visited: A, B, C, D, E, F, G, H



DFS Traversal

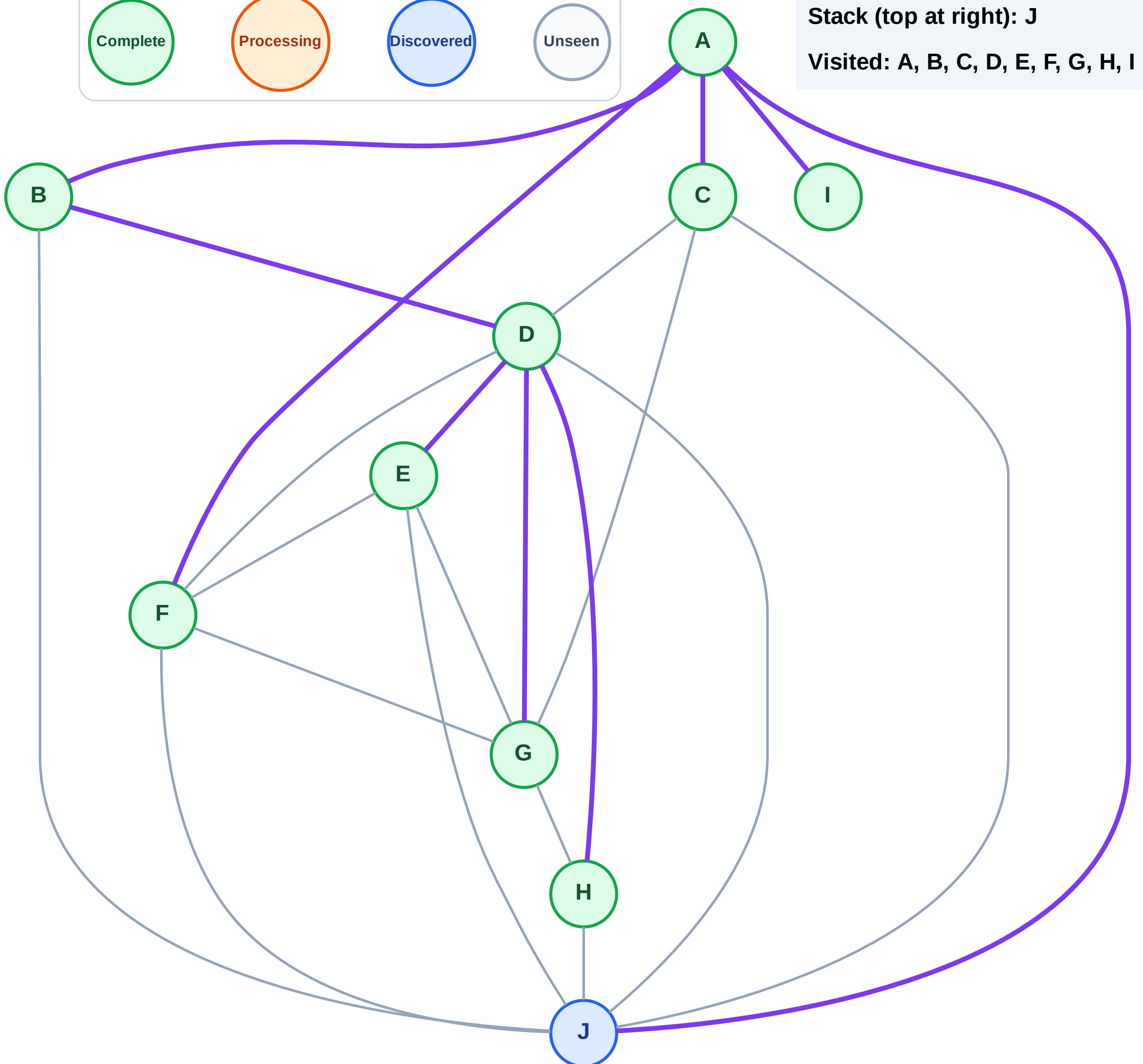
Step 19: No new neighbors from I

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

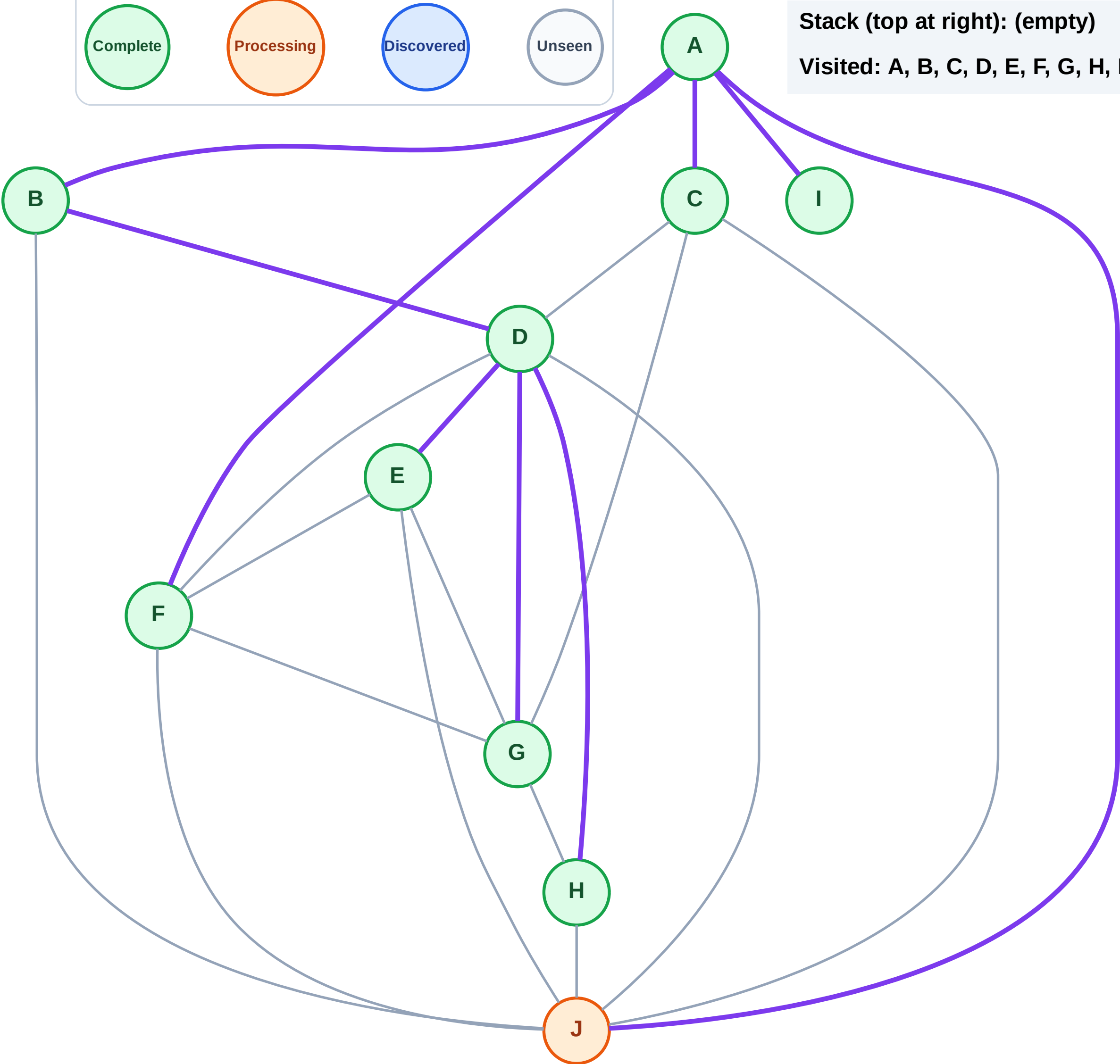
Complete

Processing

Discovered

Unseen

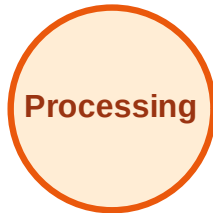
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

